PUBLISHED: 16-FEB-2017
2014.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

POWERTRAIN CONTROL MODULE (PCM) - P043E-00 TO U2108-86 - 5.0 L PETROL

WARNING:

Fuel injector voltage will reach 65Volts during operation and have a high current requirement.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.

- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Powertrain Control Module (PCM) - 5.0L Petrol. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.
For additional information, refer to: Electronic Engine Controls (303-14 Electronic Engine Controls - V8 5.0L Petrol, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P043E-00	Evaporative Emission System Leak Detection Reference Orifice Low Flow - No sub type information	▪ Diagnostic module tank leakage module internal failure	NOTES: ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ If evaporative emissions system leak related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Install a new diagnostic module tank leakage module as necessary
P043F-00	Evaporative Emission System Leak Detection Reference Orifice High Flow - No sub type information	▪ Diagnostic module tank leakage module internal failure	NOTES: ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ If evaporative emissions system leak related DTCs are also set, perform the relevant corrective action(s) first.

			▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Install a new diagnostic module tank leakage module as necessary
P0442-00	Evaporative Emission System Leak Detected (small leak) - No sub type information	▪ Evaporative emissions system leak	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the relevant section of the workshop manual and check the evaporative emissions system for leaks.
P0444-00	Evaporative Emission System Purge Control Valve A Circuit Open - No sub type information	▪ Purge valve circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the purge valve circuit for open circuit, high resistance
P0456-00	Evaporative Emission System Leak Detected (very small leak) - No sub type information	▪ Evaporative emissions system leak	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the

			customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the relevant section of the workshop manual and check the evaporative emissions system for leaks.
P0457-76	Evaporative Emission System Leak Detected (fuel cap loose/off) - Wrong mounting position	▪ Evaporative emissions system leak	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the relevant section of the workshop manual and check the evaporative emissions system for leaks.
P0458-00	Evaporative Emission System Purge Control Valve Circuit Low - No sub type information	▪ Purge valve circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to ground
P0459-00	Evaporative Emission System Purge Control Valve Circuit High - No sub type information	▪ Purge valve circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to power
P0460-29	Fuel Level Sensor A Circuit - Signal invalid	**NOTE:** **Monitor description.** Open circuit detected by the instrument cluster of the active fuel level sensor. The	▪ Check connector is not disconnected, connector pin is not backed out ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored

		instrument cluster will transmit the failure over CAN to the engine control module. The engine control module sets the DTC - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel level sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Fuel level sensor failure	DTCs using the 'Diagnosis Menu' tab and retest - Check and install new fuel level sensor as required
P0461-00	Fuel Level Sensor A Circuit Range /Performance - No sub type information	NOTE: **Monitor description.** Within the engine control module software a fuel level model is created. If the error between the fuel level model and the measured fuel level has gone over a threshold, and no change in fuel level is detected a DTC is set - The fuel tank active side fuel level sensor is stuck	- Vehicle Conditions to enable DTC Logging strategy - Consumption of 20 litres of fuel - Prioritised Checks to Perform - Check the fuel level sensor for correct movement across the entire range
P0480-11	Fan 1 Control Circuit - Circuit short to ground	- Connector is disconnected, connector pin is backed out, connector pin corrosion	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		▪ Engine cooling fan circuit short circuit to ground ▪ Engine cooling fan failure	▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine cooling fan as required
P0480-12	Fan 1 Control Circuit - Circuit short to battery	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan circuit short circuit to power ▪ Engine cooling fan failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine cooling fan as required
P0480-13	Fan 1 Control Circuit - Circuit open	NOTE: **Monitor description.** Circuit check ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan circuit open circuit, high resistance ▪ Engine cooling fan failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine cooling fan as required
P0481-11	Fan 2 Control Circuit - Circuit short to ground	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan circuit short circuit to ground ▪ Engine cooling fan failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine cooling fan as required

P0481-12	Fan 2 Control Circuit - Circuit short to battery	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine cooling fan circuit short circuit to power - Engine cooling fan failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine cooling fan as required
P0481-13	Fan 2 Control Circuit - Circuit open	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Engine cooling fan circuit open circuit, high resistance - Engine cooling fan failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check engine cooling fan circuit for open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine cooling fan as required
P0483-36	Fan Performance - Signal frequency too low	- Engine cooling fan **partially** stalled - Engine cooling fan stalling caused by deep water wading - Engine cooling fan stalling caused by obstruction in fan cowling - Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Confirm if customer has been deep water wading - Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required - Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0483-37	Fan Performance - Signal frequency too high	- Engine cooling fan failure	- Check and install new engine cooling fan as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0493-00	Fan Overspeed (clutch locked) - No sub type	- Engine cooling fan **partially** stalled	- Confirm if customer has been deep water wading

	information	▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by obstruction in fan cowling ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0496-00	Evaporative Emission System High Purge Flow - No sub type information	▪ Purge valve hose leaking or disconnected ▪ Purge valve circuit short circuit to power ▪ Purge valve stuck open	▪ Check the integrity of the purge valve hose ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Purge Valve Self Test. Install a new purge valve as necessary
P0497-00	Evaporative Emission System Low Purge Flow - No sub type information	▪ Purge valve hose blocked or restricted ▪ Purge valve circuit open circuit, high resistance ▪ Purge valve stuck closed	▪ Check the integrity of the purge valve hose ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Purge Valve Self Test. Install a new purge valve as necessary
P0500-31	Vehicle Speed Sensor A - No Signal	▪ Signal error for vehicle speed over CAN	▪ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for DTCs and refer to the relevant DTC index
P0501-00	Vehicle Speed Sensor A Range /Performance - No sub type information	▪ Anti-lock brake system control module fault ▪ Crankshaft position sensor damaged ▪ Blocked or damaged air intake system and mass air flow sensors	▪ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module, for related DTCs and refer to the relevant DTC index ▪ Inspect crankshaft position sensor for signs of damage and/or corrosion ▪ Check for damage to or blockages in air intake system and mass air flow sensors. Rectify as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P0501-86	Vehicle Speed Sensor "A" Circuit Range /Performance - Signal invalid	▪ Powertrain control module has received an error within the vehicle speed signal over the CAN bus ▪ Anti-lock brake system control module fault	▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
P0502-26	Vehicle Speed Sensor "A" Circuit Low - Signal rate of change below threshold	▪ The signal changes more slowly than is allowed ▪ Vehicle speed - range performance	▪ Using the Jaguar Land Rover approved diagnostic equipment check anti-lock brake system control module for related DTCs and refer to the relevant DTC index
P0504-27	Brake Switch A / B Correlation - Signal rate of change above threshold	▪ Brake pedal switch plunger failure ▪ Brake pedal switch not fitted correctly ▪ Brake pedal switch failure	▪ Check for related brake pressure DTCs within the anti-lock braking control module ▪ Check brake pedal switch is fitted correctly. Clear DTC, start the engine and press the brake pedal, using maximum travel, for greater than 1 minute taking care not to press the accelerator pedal ▪ Check and install new brake pedal switch as required
P0504-62	Brake Switch A / B Correlation - Signal compare failure	▪ Error check for brake pedal switch plausibility ▪ Brake pedal switch circuit short circuit to power, open circuit	▪ Refer to the electrical circuit diagrams and check brake pedal switch circuit for short circuit to power, open circuit. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0505-00	Idle Control System - No sub type information	**NOTE:** **Monitor description.** Engine idle oscillation greater than maximum deviation allowed during catalyst operation ▪ Intake air system blockage ▪ Front end accessory drive overload, failed or seized component	▪ Check air filter is not blocked ▪ Check the intake air system is free from restrictions ▪ Check front end accessory drive belt and driven components for overload failure or seized ▪ Check power steering pump for overload, failed or seized component

		■ Power steering pump overload, failed or seized component	
P0505-27	Idle Control System - Signal rate of change above threshold	NOTE: **Monitor description.** Engine idle oscillation greater than maximum deviation allowed during normal (warm) operation ■ Intake air system blockage ■ Front end accessory drive overload, failed or seized component ■ Power steering pump overload, failed or seized component	■ Check air filter is not blocked ■ Check the intake air system is free from restrictions ■ Check front end accessory drive belt and driven components for overload failure or seized ■ Check power steering pump for overload, failed or seized component
P0506-00	Idle Control System - RPM Lower Than Expected - No sub type information	NOTE: **Monitor description.** Engine idle speed lower than minimum deviation allowed during normal (warm) operation ■ Intake air system blockage ■ Front end accessory drive overload, failed or seized component ■ Power steering pump overload, failed or seized component	■ Check air filter is not blocked ■ Check the intake air system is free from restrictions ■ Check front end accessory drive belt and driven components for overload failure or seized ■ Check power steering pump for overload, failed or seized component
P0506-24	Idle Control System - RPM	NOTE:	■ Check air filter is not blocked

	Lower Than Expected - Signal stuck high	**Monitor description.** Engine idle speed lower than minimum deviation allowed during catalyst operation - Intake air system blockage - Front end accessory drive overload, failed or seized component - Power steering pump overload, failed or seized component	- Check the intake air system is free from restrictions - Check front end accessory drive belt and driven components for overload failure or seized - Check power steering pump for overload, failed or seized component
P0507-00	Idle Control System - RPM Higher Than Expected - No sub type information	NOTE: **Monitor description.** Engine idle speed greater than maximum deviation allowed during normal (warm) operation - Intake air system leakage - Intake air system components incorrectly installed, loose or damaged pipework - Engine crankcase breather system leakage - Engine crankcase breather system components incorrectly installed	- Using the manufacturer approved smoke test system check the intake air system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage - Check Engine crankcase breather system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage
P0507-23	Idle Control System - RPM Higher Than Expected - Signal stuck low	NOTE: **Monitor description.** Engine idle	- Using the manufacturer approved smoke test system check the intake air system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage

			speed greater than maximum deviation allowed during catalyst operation - Intake air system leakage - Intake air system components incorrectly installed, loose or damaged pipework - Engine crankcase breather system leakage - Engine crankcase breather system components incorrectly installed	- Check Engine crankcase breather system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage
	P050B-92	Cold Start Ignition Timing Performance - Performance or incorrect operation	NOTE: **Monitor description.** Actual ignition timing different to ignition timing requested for catalyst heating; off idle - The engine control module has detected that the actual ignition timing is different to that requested for optimal catalyst heating - Intake air system components incorrectly installed, loose or damaged pipework - Engine crankcase breather system leakage - Engine crankcase breather system components incorrectly installed	- Using the manufacturer approved smoke test system check the intake air system is free from leakage, components are correctly installed, pipework is not loose or damaged - Check engine vacuum system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged - Check engine crankcase breather system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged - Check air filter(s) are not blocked - Check the intake air system is free from restrictions - Check front end accessory drive belt and driven components for failure

		▪ Engine vacuum system leak ▪ Brake booster leak ▪ Engine speed error ▪ Incorrect accessory loading ▪ Engine misfire	
P050B-93	Cold Start Ignition Timing Performance - No operation	NOTE: **Monitor description.** Actual ignition timing different to ignition timing requested for catalyst heating; stationary idle ▪ The engine control module has detected that the actual ignition timing is different to that requested for optimal catalyst heating ▪ Intake air system leakage ▪ Intake air system components incorrectly installed, loose or damaged pipework ▪ Engine crankcase breather system leakage ▪ Engine crankcase breather system components incorrectly installed ▪ Engine vacuum system leak ▪ Brake booster leak ▪ Engine speed error ▪ Incorrect accessory loading ▪ Engine misfire	▪ Using the manufacturer approved smoke test system check the intake air system is free from leakage, components are correctly installed, pipework is not loose or damaged ▪ Check engine vacuum system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged ▪ Check engine crankcase breather system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged ▪ Check air filter(s) are not blocked ▪ Check the intake air system is free from restrictions ▪ Check front end accessory drive belt and driven components for failure
P050F-01	Brake Assist Vacuum Too	NOTE:	▪ Check brake vacuum is available. Deplete vacuum with engine off until the

	Low - General Electrical Failure	**Monitor description.** The engine control module checks there is brake vacuum greater than 60kPa gauge pressure after 45 seconds. After engine is running, the atmospheric pressure is greater than 75kPa, the coolant is greater than 20'C, and the brake pedal is released - Brake vacuum sensor open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Brake vacuum pump air leakage - Brake vacuum pump failure	brake pedal goes hard. With brake pedal still applied start engine, if the pedal drops then vacuum is available - Refer to the electrical circuit diagrams and check brake vacuum sensor circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - If the brake pedal does not drop check under bonnet for air leaks and brake vacuum pump operation - Check and install new brake vacuum pump as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0512-12	Starter Request Circuit - Circuit short to battery	NOTE: **Monitor description.** The engine control module checks that the hard wired crank request from the body control module matches the powermode CAN signal - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check starter request circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		- Starter request circuit short circuit to power	
P0512-14	Starter Request Circuit - Circuit short to ground or open	NOTE: **Monitor description.** The engine control module checks that the hard wired crank request from the body control module matches the powermode CAN signal - Connector is disconnected, connector pin is backed out, connector pin corrosion - Starter request circuit open circuit, short circuit to ground	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check starter request circuit for open circuit, short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0512-72	Starter Request Circuit - Actuator stuck open	- Starter relay stuck open - Connector is disconnected, connector pin is backed out, connector pin corrosion - Starter request diagnostic circuit open circuit - Starter relay failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - If engine starts turning, refer to the electrical circuit diagrams and check starter request circuit for open circuit - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new start relay as required
P0512-73	Starter Request Circuit - Actuator stuck closed	- Starter relay stuck closed - Connector is disconnected, connector pin is backed out, connector pin corrosion - Starter request diagnostic circuit short circuit to power - Starter relay failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - If engine starts turning, refer to the electrical circuit diagrams and check starter request circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

			▪ Check and install new start relay as required
P0513-00	Incorrect Immobilizer Key - No sub type information	▪ Invalid signal from central junction box immobilizer	▪ Check for CAN network and engine control module related DTCs and refer to the relevant DTC index. ▪ Check the central junction box for related DTCs and refer to the relevant DTC index. ▪ Re-configure the engine control module using the Jaguar Land Rover approved diagnostic equipment ▪ Re-configure the central junction box using the Jaguar Land Rover approved diagnostic equipment
P0528-00	Fan Speed Sensor Circuit No Signal - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan stalling caused by obstruction in fan cowling ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Engine cooling fan failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install new engine cooling fan as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P052A-00	Cold Start Intake (A) Camshaft Position Timing Over-Advanced (Bank 1) - No sub type information	NOTE: **Monitor description.** Inlet camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase ▪ Oil level is low ▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped	▪ Check the oil level is correct ▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped ▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable camshaft timing actuator as required

		▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure is low ▪ Variable camshaft timing actuator mechanical failure ▪ Excessive camshaft friction, restricted movement or stuck	▪ Refer to the relevant sections of the workshop manual and check the camshaft operation
P052C-00	Cold Start Intake (A) Camshaft Position Timing Over-Advanced (Bank 2) - No sub type information	NOTE: **Monitor description.** Inlet camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase ▪ Oil level is low ▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped ▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure is low ▪ Variable camshaft timing actuator mechanical failure ▪ Excessive camshaft friction, restricted movement or stuck	▪ Check the oil level is correct ▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped ▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable camshaft timing actuator as required ▪ Refer to the relevant sections of the workshop manual and check the camshaft operation
P053F-23	Cold Start Fuel Pressure Performance - Signal stuck low	NOTE: **Monitor description.** The high	▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system

		pressure fuel pump signal is below a threshold during catalyst heating ▪ External fuel leak from high pressure fuel system ▪ Internal fuel leak into low pressure fuel system	
P053F-24	Cold Start Fuel Pressure Performance - Signal stuck high	NOTE: **Monitor description.** The high pressure fuel pump signal is above a threshold during catalyst heating ▪ External fuel leak from high pressure fuel system ▪ Internal fuel leak into low pressure fuel system	▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system
P054A-00	Cold Start Exhaust (B) Camshaft Position Timing Over-Advanced (Bank 1) - No sub type information	NOTE: **Monitor description.** Exhaust camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase ▪ Oil level is low ▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped	▪ Check the oil level is correct ▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped ▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable camshaft timing actuator as required

		■ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Oil pressure is low ■ Variable camshaft timing actuator mechanical failure ■ Excessive camshaft friction, restricted movement or stuck	■ Refer to the relevant sections of the workshop manual and check the camshaft operation
P054C-00	Cold Start Exhaust (B) Camshaft Position Timing Over-Advanced (Bank 2) - No sub type information	NOTE: **Monitor description.** Exhaust camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase ■ Oil level is low ■ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped ■ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Oil pressure is low ■ Variable camshaft timing actuator mechanical failure ■ Excessive camshaft friction, restricted movement or stuck	■ Check the oil level is correct ■ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped ■ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing actuator as required ■ Refer to the relevant sections of the workshop manual and check the camshaft operation
P0561-00	System Voltage Unstable - No sub type information	■ Power and ground connections to the engine control module, open circuit, high resistance	■ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests

		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Check the vehicle charging system performance to ensure the voltage regulation is correct
P0562-00	System Voltage Low - No sub type information	▪ Vehicle battery under charged ▪ Charging system fault - under charging	▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Check the vehicle charging system performance to ensure the voltage regulation is correct
P0562-16	System Voltage Low - Circuit voltage below threshold	**NOTE:** **Monitor description.** Engine control module internal 5 Volt supply below the minimum hardcoded threshold value ▪ Power and ground connections to the engine control module, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0563-00	System Voltage High - No sub type information	▪ Vehicle battery over charged ▪ Charging system fault - over charging	▪ Check the vehicle charging system performance to ensure the voltage regulation is correct

P0563-17	System Voltage High - Circuit voltage above threshold	NOTE: **Monitor description.** Engine control module internal 5 Volt supply above the upper hardcoded threshold value - Charging system fault - over charging - Engine control module failure	- Check the vehicle charging system performance to ensure the voltage regulation is correct - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0571-62	Brake Switch A Circuit - Signal compare failure	- The brake pedal switch signal received over CAN is defective - Brake pedal switch detached from pedal box whilst electrically connected - Brake pedal switch adjustment incorrect - Brake pedal switch harness failure - Brake pedal switch failure	- Using the Jaguar Land Rover approved diagnostic equipment carry out network integrity test - Check for related DTCs and refer to the relevant DTC index - Inspect brake switch connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the brake pedal switch circuits for open circuit, short circuit to ground, short circuit to power - Check and install new brake pedal switch as required
P0575-81	Cruise Control Input Circuit - Invalid serial data received	- The DTC sets if the cancel, set minus, set plus, resume, headway increase and headway decrease speed control buttons have been pressed for longer than a calibrated period of time. The system then assumes a stuck/damaged button and will cancel and/or disable cruise. The failure will be healed in the next driving cycle if the failure is removed	- Check speed control buttons are not jammed/contaminated/damaged. Check speed control module for related DTCs and refer to the relevant DTC index - Check steering wheel clock spring and connections to button pack for damage - Check and install new cruise control button pack as required
P0597-13	Thermostat Heater Control Circuit / Open	- The engine control module has determined an open	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

	- Circuit open	circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Coolant thermostat heater circuit, open circuit, high resistance ▪ Coolant thermostat heater failure	▪ Refer to the electrical circuit diagrams and check coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new coolant thermostat heater as required
P0597-4B	Thermostat Heater Control Circuit/Open - Over temperature	▪ The powertrain control module detected an internal temperature above the expected range ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Coolant thermostat heater circuit, open circuit, high resistance ▪ Circuit reference - O_T_CTH - ▪ Coolant thermostat heater failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new coolant thermostat heater as required
P0598-11	Thermostat Heater Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Coolant thermostat heater circuit, short circuit to ground	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check coolant thermostat heater circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new coolant thermostat heater as required

		▪ Coolant thermostat heater failure	
P0599-12	Thermostat Heater Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Coolant thermostat heater circuit, short circuit to power ▪ Coolant thermostat heater failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check coolant thermostat heater circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new coolant thermostat heater as required
P05A0-97	Active Grille Air Shutter "A" Stuck On - Component or system operation obstructed or blocked	▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Active grille air shutter circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Active grille air shutter unable to move due to obstruction ▪ Active grille air shutter failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check there is not any obstruction preventing active grille air shutter movement ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new active grille air shutter as required
P05A2-01	Active Grille Air Shutter "A" Control Circuit /Open - General electrical failure	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the active grille air shutter

		- Active grille air shutter circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Active grille air shutter unable to move due to obstruction - Active grille air shutter failure	circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check there is not any obstruction preventing active grille air shutter movement - Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest - Check and install a new active grille air shutter as required
P05A2-97	Active Grille Air Shutter "A" Control Circuit /Open - Component or system operation obstructed or blocked	- The powertrain control module has detected that the operation of a component is prevented by an obstruction - Other related DTCs - Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Active grille air shutter unable to move due to obstruction - Active grille air shutter failure	- Check for related DTCs and refer to the relevant DTC index - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check there is not any obstruction preventing active grille air shutter movement - Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest - Check and install a new active grille air shutter as required
P05A6-01	Active Grille Air Shutter "A" Supply Voltage Circuit/Open - General electrical failure	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Active grille air shutter circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Active grille air shutter unable to move due to obstruction - Active grille air shutter failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check there is not any obstruction preventing active grille air shutter movement - Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest - Check and install a new active grille air shutter as required

P05C0-4B	Active Grille Air Shutter Module "A" Over Temperature - Over temperature	▪ The powertrain control module detected an internal temperature above the expected range ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Active grille air shutter circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Active grille air shutter unable to move due to obstruction ▪ Active grille air shutter failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check there is not any obstruction preventing active grille air shutter movement ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new active grille air shutter as required
P0606-00	Control Module Processor - No sub type information	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-02	Control Module Processor - General signal failure	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-17	Control Module Processor - Circuit voltage above threshold	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-48	Control Module Processor - Supervision software failure	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-	Control Module		

86	Processor - Signal invalid	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-92	Control Module Processor - Performance or incorrect operation	NOTE: **Monitor description.** DTC is set if a software reset has occurred for an abnormal reason ▪ Software reset has occurred ▪ Low battery reset during start attempt	▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Clear the DTC and test drive the vehicle including at least five ignition cycles. Check for DTCs. If DTC is not present return to customer. If DTC repeatedly returns check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0606-94	Control Module Processor - Unexpected operation	NOTE: **Monitor description.** DTC is set if a software reset has occurred for an abnormal reason ▪ Software reset has occurred	▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Clear the DTC and test drive the vehicle including at least five ignition cycles. Check for DTCs. If DTC is not present return to customer. If DTC repeatedly returns check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P060A-00	Internal Control Module Monitoring Processor Performance - No sub type information	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P060B-00	Internal Control Module A/D Processing Performance - No sub type information	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"

P060B-02	Internal Control Module A/D Processing Performance - General signal failure	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P060C-00	Internal Control Module Main Processor Performance - No sub type information	▪ Engine speed calculation not plausible. Engine control module software error ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P060D-00	Internal Control Module Accelerator Pedal Position Performance - No sub type information	▪ Monitoring level 1 accelerator pedal diagnostics have not captured a synchronisation fault of the accelerator pedal. Engine control module software error ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0615-04	Starter Relay Circuit - System Internal Failures	▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"

P0615-13	Starter Relay Circuit - Circuit open	▪ Starter motor relay control circuit open circuit ▪ Starter motor relay disconnected/missing ▪ Starter motor relay failure	▪ Refer to the electrical circuit diagrams and check starter motor relay control circuit for open circuit ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
P0615-72	Starter Relay Circuit - Actuator stuck open	▪ Starter motor fuse has blown ▪ Starter motor relay is stuck open ▪ Starter motor relay control circuit failure	▪ Check starter motor fuse ▪ Refer to the electrical circuit diagrams and check starter motor relay control circuit for open circuit, short circuit to ground, short circuit to power, high resistance ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
P0615-81	Starter Relay Circuit - Invalid serial data received	▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0616-00	Starter Relay Circuit Low - No sub type information	▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest

P0616-11	Starter Relay Circuit Low - Circuit short to ground	▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0617-00	Starter Relay Circuit High - No sub type information	▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0617-12	Starter Relay Circuit High - Circuit short to battery	▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to power ▪ Check and install new starter motor relay as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P061A-00	Internal Control Module Torque Performance - No sub type information	▪ Delivered torque is calculated as being above requested torque within monitoring system. Engine control module software error ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061C-	Internal		

00	Control Module Engine RPM Performance - No sub type information	- Desired ignition angle not plausible. Engine control module software error - Engine control module failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061D-00	Internal Control Module Engine Air Mass Performance - No sub type information	- Fuel ethanol content correction factor applied by control system is not plausible to monitoring calculation. Engine control module software error - Oil evaporation content of fuel mass is calculated erroneously low within control system when compared to rational check within monitoring system. Engine control module software error - Engine control module failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061D-42	Internal Control Module Engine Air Mass Performance - General memory failure	- Injector cut off patterns found not plausible. Engine control module software error - Engine control module failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061D-43	Internal Control Module Engine Air Mass	- Desired fuelling control limits implausible. Engine control module software error	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment check and install

	Performance - Special memory failure	▪ Engine control module failure	latest relevant level of software to the engine control module ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061D-62	Internal Control Module Engine Air Mass Performance - Signal compare failure	▪ Predicted air mass was found not plausible. Engine control module software error ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P061F-00	Internal Control Module Throttle Actuator Controller Performance - No sub type information	NOTE: **Monitor description.** Engine control module monitors the measured throttle position against the requested position and sets the DTC if they deviate excessively ▪ Other throttle related DTCs ▪ Blockages/restriction preventing free motion of the throttle blade ▪ Throttle actuator failure	▪ Check engine control module for related DTCs and refer to relevant DTC index. Rectify these first ▪ Inspect throttle for signs of blockage or restriction that prevent free motion of the throttle blade ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0627-07	Fuel Pump A Control Circuit / Open - Mechanical Failures	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check low pressure fuel pump

		• Harness failure between fuel pump driver module and low pressure fuel pump • Low pressure fuel pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance • Low pressure fuel pump failure	control circuit for short circuit to ground, short circuit to power, open circuit, high resistance • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest • Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump • Check and install new low pressure fuel pump as required
P062A-31	Fuel Pump A Control Circuit Range /Performance - No signal	• Connector is disconnected, connector pin is backed out, connector pin corrosion • Fuel pump driver module signal circuit short circuit to ground, short circuit to power, open circuit, high resistance • Fuel pump driver module failure	• Inspect connectors for signs of water ingress, and pins for damage and/or corrosion • Refer to the electrical circuit diagrams and check fuel pump driver module signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest • Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump • Check and install new fuel pump driver module as required
P062F-00	Internal Control Module EEPROM Error - No sub type information	**NOTE:** **Monitor description.** Engine control module monitors for hardware error of the EEPROM and sets the DTC when at least three blocks cannot be read • Engine control module failure	• Check and install new engine control module as required • Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P062F-43	Internal Control Module	**NOTE:**	• Check and install new engine control module as required

	EEPROM Error - Special memory failure	**Monitor description.** Engine control module monitors for hardware error of the EEPROM and sets the DTC when one block can not be written more than 3 times - Engine control module failure	- Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P062F-96	Internal Control Module EEPROM Error - Component internal failure	NOTE: **Monitor description.** Engine control module monitors for hardware error of the EEPROM and sets the DTC if the sector erase cannot be performed or successfully completed - Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0633-00	Immobilizer Key Not Programmed - ECM/PCM - No sub type information	- Power is lost from the engine control module or the central junction box during the immobilizer learn routine	- Refer to the electrical circuit diagrams and check the power and ground connections to the engine control module and central junction box - Using the Jaguar Land Rover approved diagnostic equipment, carry out the immobilisation procedure - Check for CAN network interference /engine control module related errors - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module
P0633-42	Immobilizer Key Not Programmed -	- Security key not programmed - Engine control module failure	- Using the Jaguar Land Rover approved diagnostic equipment, carry out the immobilisation procedure

	ECM/PCM - General memory failure		- Check for CAN network interference /engine control module related errors. Re-configure the engine control module using the Jaguar Land Rover approved diagnostic equipment. Re-configure the instrument panel control module using the Jaguar Land Rover approved diagnostic equipment - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0634-00	Control Module Internal Temperature "A" Too High - No sub type information	NOTE: **Monitor description.** The internal temperature of the engine control module is above the specified limit - Engine control module internal temperature too high - Engine control module cooling fan failure - Engine control module failure	- Check engine control module for related DTCs and refer to relevant DTC index. Rectify these first - Check the engine control module cooling fan is operational - Check and install new engine control module cooling fan as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0642-00	Sensor Reference Voltage A Circuit Low - No sub type information	NOTE: **Monitor description.** Engine control module internal 5 Volt supply below the minimum hardcoded threshold value Power and ground connections to the	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine control module as required

		engine control module, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine control module failure	▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0643-00	Sensor Reference Voltage A Circuit High - No sub type information	NOTE: **Monitor description.** Engine control module internal 5 Volt supply above the upper hardcoded threshold value ▪ Charging system fault - over charging ▪ Engine control module failure	▪ Check the vehicle charging system performance to ensure the voltage regulation is correct ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064A-38	Fuel Pump Control Module - Signal frequency incorrect	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pump driver module diagnostic line circuit open circuit, short circuit to power, short circuit to ground ▪ Fuel pump driver module failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel pump driver module diagnostic circuit for open circuit, short circuit to power, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump ▪ Check and install new fuel pump driver module as required
P064A-42	Fuel Pump Control Module "A" - General memory failure	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel pump driver module

		▪ Fuel pump driver module diagnostic line circuit open circuit, short circuit to power, short circuit to ground ▪ Fuel pump driver module failure	diagnostic circuit for open circuit, short circuit to power, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new fuel pump driver module as required
P064A-98	Fuel Pump Control Module "A" - Component or system over temperature	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pump driver module diagnostic line circuit open circuit, short circuit to power, short circuit to ground ▪ Fuel pump driver module failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel pump driver module diagnostic circuit for open circuit, short circuit to power, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new fuel pump driver module as required
P064D-08	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Bus Signal / Message Failures	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064D-16	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Circuit voltage below threshold	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064D-17	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Circuit voltage above threshold	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064D-86	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Signal invalid	▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064E-	Internal		

08	Control Module O2 Sensor Processor Performance - Bank 2 - Bus Signal / Message Failures	- Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064E-16	Internal Control Module O2 Sensor Processor Performance - Bank 2 - Circuit voltage below threshold	- Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064E-17	Internal Control Module O2 Sensor Processor Performance - Bank 2 - Circuit voltage above threshold	- Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P064E-86	Internal Control Module O2 Sensor Processor Performance - Bank 2 - Signal invalid	- Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0652-00	Sensor Reference Voltage B Circuit Low - No sub type information	NOTE: **Monitor description.** Engine control module internal 5 Volt supply below the minimum hardcoded threshold value - Power and ground connections to the engine control module, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"

		▪ Engine control module failure	
P0653-00	Sensor Reference Voltage B Circuit High - No sub type information	NOTE: **Monitor description.** Engine control module internal 5 Volt supply above the upper hardcoded threshold value ▪ Charging system fault - over charging ▪ Engine control module failure	▪ Check the vehicle charging system performance to ensure the voltage regulation is correct ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0657-12	Actuator Supply Voltage A Circuit / Open - Circuit short to battery	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required
P0657-13	Actuator Supply Voltage A Circuit Open - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required

P0658-11	Actuator Supply Voltage A Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required
P0658-13	Actuator Supply Voltage A Circuit Low - Circuit open	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required
P0659-11	Actuator Supply Voltage A Circuit High - Circuit short to ground	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required
P0659-12	Actuator Supply Voltage A Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored

		power measurement when another value was expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity ▪ Sound symposer failure	DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new sound symposer as required
P0660-00	Intake Manifold Tuning Valve Control Circuit / Open - Bank 1 - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Intake manifold tuning valve control circuit open circuit ▪ Intake manifold tuning valve sensor failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check intake manifold tuning valve control circuit for open circuit, short circuit to power, short circuit to ground, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new intake manifold tuning valve sensor as required
P0661-00	Intake Manifold Tuning Valve Control Circuit Low - Bank 1 - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Intake manifold tuning valve control circuit open circuit, short circuit to ground, high resistance ▪ Intake manifold tuning valve sensor failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check intake manifold tuning valve control circuit for open circuit, short circuit to power, short circuit to ground, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new intake manifold tuning valve sensor as required
P0662-00	Intake Manifold Tuning Valve Control Circuit High - Bank 1 - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Intake manifold tuning valve control circuit short circuit to power, high resistance	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check intake manifold tuning valve control circuit for open circuit, short circuit to power, short circuit to ground, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Intake manifold tuning valve sensor failure	▪ Check and install new intake manifold tuning valve sensor as required
P0666-00	Control Module Internal Temperature Sensor "A" Circuit - No sub type information	▪ Engine control module internal communication error ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0666-11	Control Module Internal Temperature Sensor "A" Circuit - Circuit short to ground	▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Powertrain control module internal temperature sensor failure ▪ Powertrain control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P0666-12	Control Module Internal Temperature Sensor "A" Circuit - Circuit short to battery	▪ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Powertrain control module internal temperature sensor failure ▪ Powertrain control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P0666-64	Control Module Internal Temperature Sensor "A"	▪ Powertrain control module internal temperature sensor failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module

	Circuit - Signal plausibility failure	▪ Powertrain control module failure	▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P0667-00	Control Module Internal Temperature Sensor "A" Range /Performance - No sub type information	▪ Engine control module internal temperature sensor less than specified limit ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0667-84	Control Module Internal Temperature Sensor "A" Range /Performance - Signal below allowable range	▪ Engine control module internal temperature sensor less than specified limit ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0667-85	Control Module Internal Temperature Sensor "A" Range /Performance - Signal above allowable range	▪ Engine control module internal temperature sensor greater than specified limit ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P0686-11	ECM/PCM Power Relay	▪ Engine control module power relay circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check engine control module power relay circuit for short circuit to ground

	Control Circuit Low - Circuit short to ground		■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0687-24	ECM/PCM Power Relay Control Circuit High - Signal stuck high	■ Engine control module power relay circuit short circuit to power, short circuit to another circuit	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Refer to the electrical circuit diagrams and check engine control module power relay circuit for short circuit to power, short circuit to another circuit
P0688-00	ECM/PCM Power Relay Sense Circuit - No sub type information	■ Engine control module supply voltage greater than 16 volts ■ Charging system fault - over charging ■ Use of battery charger with boost setting producing greater than 16 volts ■ Use of 24 volt charging or jump starting system	■ Check the vehicle charging system performance to ensure the voltage regulation is correct ■ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual
P068A-00	ECM/PCM Power Relay De-Energized - Too Early - No sub type information	■ Engine control module main power relay de-energised before normal shutdown processes ■ Power and ground connections to the engine control module, open circuit, high resistance ■ Battery disconnection or relay control circuit intermittent harness failure ■ Engine control module power relay failure	■ This DTC is set when the engine management system high current relay contacts open early - Indicating a power hold failure. Check the vehicle battery has not been disconnected before the engine management system relay has powered down. Check the operation of the engine management system high current relay. Refer to the electrical circuit diagrams and check the engine management system high current relay supply and control circuits for open circuits, high resistance, short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Clear DTC and retest ■ Check and install new relay as required
P0691-11	Fan 1 Control Circuit Low - Circuit short to ground	■ Engine cooling fan circuit for short circuit to ground ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to ground ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install new engine cooling fan as required

		▪ Engine cooling fan failure	▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0692-12	Fan 1 Control Circuit High - Circuit short to battery	▪ Engine cooling fan circuit for short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine cooling fan failure	▪ Refer to the electrical circuit diagrams and check engine cooling fan circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install new engine cooling fan as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P06A6-00	Sensor Reference Voltage A Circuit Range /Performance - No sub type information	▪ Engine control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to electrical circuit diagrams and check the engine control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
P06A7-00	Sensor Reference Voltage B Circuit Range /Performance - No sub type information	▪ Engine control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to electrical circuit diagrams and check the engine control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
P06A8-00	Sensor Reference Voltage C Circuit Range /Performance - No sub type information	▪ Engine control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit ▪ Connector is disconnected, connector pin is	▪ Refer to electrical circuit diagrams and check the engine control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		backed out, connector pin corrosion	
P06DA-00	Engine Oil Pressure Control Circuit /Open - No sub type information	▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, open circuit, high resistance ▪ Variable oil pump failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new variable oil pump as required
P06DA-13	Engine Oil Pressure Control Circuit / Open - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, open circuit, high resistance ▪ Variable oil pump failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable oil pump as required
P06DB-00	Engine Oil Pressure Control Circuit Low - No sub type information	▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, short circuit to ground ▪ Variable oil pump failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new variable oil pump as required
P06DB-11	Engine Oil Pressure Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		detected a ground measurement when another value was expected ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, short circuit to ground ▪ Variable oil pump failure	▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable oil pump as required
P06DC-00	Engine Oil Pressure Control Circuit High - No sub type information	▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, short circuit to power ▪ Variable oil pump failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ▪ Check and install a new variable oil pump as required
P06DC-12	Engine Oil Pressure Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ▪ Variable oil pump circuit, short circuit to power ▪ Variable oil pump failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check variable oil pump circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new variable oil pump as required
P06E9-07	Sensor Power Supply "C" Circuit High - Mechanical Failures	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check engine control module for additional DTCs and refer to the relevant DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		■ Harness failure - Wiring integrity	■ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P0700-00	Transmission Control System (MIL Request) - No sub type information	■ MIL request by automatic gearbox ■ HS CAN network, short circuit to power, short circuit to ground, open circuit	■ Check transmission control module for DTCs and refer to the relevant DTC index. Clear DTC and retest ■ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit
P0703-62	Brake Switch B Circuit - Signal compare failure	■ The brake pedal switch signal received over CAN is defective ■ Brake pedal switch detached from pedal box whilst electrically connected ■ Brake pedal switch adjustment incorrect ■ Brake pedal switch harness failure ■ Brake pedal switch failure	■ Using the Jaguar Land Rover approved diagnostic equipment carry out network integrity test ■ Check for related DTCs and refer to the relevant DTC index ■ Inspect brake switch connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check the brake pedal switch circuits for open circuit, short circuit to ground, short circuit to power ■ Check and install new brake pedal switch as required
P0817-13	Starter Disable Circuit / Open - Circuit open	■ Starter motor pinion relay failure ■ Starter motor pinion relay circuit short circuit open circuit, high resistance	■ Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for open circuit, high resistance ■ Check and install new starter motor pinion relay as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081A-11	Starter Disable Circuit Low - Circuit short to ground	■ Starter motor pinion relay failure ■ Starter motor pinion relay circuit short circuit to ground	■ Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to ground ■ Check and install new starter motor pinion relay as required ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081A-12	Starter Disable Circuit Low -	■ Starter motor pinion relay failure	

	Circuit short to battery	• Starter motor pinion relay circuit short circuit to power	• Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to power • Check and install new starter motor pinion relay as required • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081A-4B	Starter Disable Circuit Low - Over temperature	• Starter motor pinion relay failure • Starter motor pinion relay circuit short circuit to ground, short circuit to power, open circuit, high resistance • Engine control module failure	• Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance • Check and install new starter motor pinion relay as required • Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module • Check and install new engine control module as required • Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module" • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081B-11	Starter Disable Circuit High - Circuit short to ground	• Starter motor pinion relay failure • Starter motor pinion relay circuit short circuit to ground	• Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to ground • Check and install new starter motor pinion relay as required • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081B-12	Starter Disable Circuit High - Circuit short to battery	• Starter motor pinion relay failure • Starter motor pinion relay circuit short circuit to power	• Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to power • Check and install new starter motor pinion relay as required • Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P081B-4B	Starter Disable Circuit High - Over	• Starter motor pinion relay failure	• Refer to the electrical circuit diagrams and check starter motor pinion relay circuit for short circuit to ground, short

	temperature	- Starter motor pinion relay circuit short circuit to ground, short circuit to power, open circuit, high resistance - Engine control module failure	circuit to power, open circuit, high resistance - Check and install new starter motor pinion relay as required - Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module" - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
P0850-00	Park / Neutral Switch Input Circuit - No sub type information	- Hardwired park /neutral switch does not match the information within the CAN signal - Park/neutral switch circuit, short circuit to power, short circuit to ground, open circuit - HS CAN network failure	- Refer to the electrical circuit diagrams and check the park/neutral switch circuit, for short circuit to power, short circuit to ground, open circuit - If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit
P0850-02	Park/Neutral Switch Input Circuit - General signal failure	- Park/neutral signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Missing/invalid data from the transmission control switch	- Refer to the electrical circuit diagrams and check the park/neutral signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related DTCs and refer to the relevant DTC index
P0850-64	Park/Neutral Switch Input Circuit - Signal plausibility failure	- Park/neutral signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Missing/invalid data from the transmission control switch	- Refer to the electrical circuit diagrams and check the park/neutral signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related DTCs and refer to the relevant DTC index
P0A0F-07	Engine Failed to Start -		

	Mechanical failures	- Car configuration file incorrect for stop /start system - Harness failure - Wiring integrity starter motor circuit - Tandem solenoid starter motor failure - Tandem solenoid starter motor pinion not operating - Harness failure - Wiring integrity camshaft sensor circuit - Harness failure - Wiring integrity crankshaft sensor circuit	- Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P1114-16	Intake Air Temperature 2 Circuit Low (Super /Turbocharged engines) - Circuit voltage below threshold	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P1115-17	Intake Air Temperature 2 Circuit High (Super /Turbocharged engines) - Circuit voltage above threshold	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P115D-00	Mass Air Flow Circuit Offset - No sub type information	NOTE: **Monitor description.** Mass air flow sensor signal failure - Blocked air filter - Other related DTCs	- Check air filter is serviceable - Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index - Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage - Check mass air flow sensor for contamination by oil or debris clean as required

		▪ Air induction system, incorrectly installed components, loose hose clips or air leakage ▪ Mass air flow sensor contamination by oil or debris ▪ Electric throttle blade contamination by oil or debris ▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Throttle adaption failure	▪ Check electric throttle blade for contamination by oil or debris clean as required ▪ Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P115D-21	Mass Air Flow Circuit Offset - Signal amplitude < minimum	**NOTE:** **Monitor description.** Mass air flow sensors left /right flow to low ▪ Blocked air filter ▪ Other related DTCs ▪ Air induction system, incorrectly installed components, loose hose clips or air leakage ▪ Mass air flow sensor contamination by oil or debris ▪ Electric throttle blade contamination by oil or debris ▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Throttle adaption failure	▪ Check air filter is serviceable ▪ Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index ▪ Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage ▪ Check mass air flow sensor for contamination by oil or debris clean as required ▪ Check electric throttle blade for contamination by oil or debris clean as required ▪ Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P115D-22	Mass Air Flow Circuit Offset -	**NOTE:**	▪ Check air filter is serviceable

		Signal amplitude > maximum	**Monitor description.** Mass air flow sensors left /right flow to high - Blocked air filter - Other related DTCs - Air induction system, incorrectly installed components, loose hose clips or air leakage - Mass air flow sensor contamination by oil or debris - Electric throttle blade contamination by oil or debris - Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Throttle adaption failure	- Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index - Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage - Check mass air flow sensor for contamination by oil or debris clean as required - Check electric throttle blade for contamination by oil or debris clean as required - Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
	P115D-64	Mass Air Flow Circuit Offset - Signal plausibility failure	NOTE: **Monitor description.** Mass air flow sensor plausibility failure - Blocked air filter - Other related DTCs - Air induction system, incorrectly installed components, loose hose clips or air leakage - Mass air flow sensor contamination by oil or debris - Electric throttle blade contamination by oil or debris	- Check air filter is serviceable - Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index - Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage - Check mass air flow sensor for contamination by oil or debris clean as required - Check electric throttle blade for contamination by oil or debris clean as required - Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		- Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Throttle adaption failure	- Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P115F-11	Electronic Control Module Cooling Fan Circuit - Circuit short to ground	- The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Powertrain control module cooling fan circuit, short circuit to ground - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity	- Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for short circuit to ground - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
P115F-13	Electronic Control Module Cooling Fan Circuit - Circuit open	- The powertrain control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Powertrain control module cooling fan circuit, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity	- Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
P1315-00	Persistent Misfire - No sub type	- Other related DTCs - Fuel level, poor fuel quality/incorrect	- Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and

	information	grade, fuel contaminated - Spark plugs damaged/worn - Incorrect position or damaged crank shaft position sensor. Reluctor ring /toothed wheel damaged - Excessive engine wear - Catalyst blocked /restriction in exhaust system - Driveline imbalance	refer to the relevant DTC index. Rectify these first - Check the fuel level, add fuel if required - Check for poor quality or contaminated fuel - Check spark plugs for poor condition and replace as necessary - Check for damaged or incorrectly positioned crank shaft position sensor, reluctor ring/toothed wheel - Check for poor cylinder compression, incorrect valve clearance and excessive engine wear - Check for driveline imbalances, front end ancillary drives and transmission mounts
P131A-00	Low Fuel Level Detection - No sub type information	**NOTE:** **Monitor description.** The engine control module sets the DTC when other DTCs are set and when the fuel level is low - Other fuel related DTCs are set - Low level fuel condition	- Using the Jaguar Land Rover approved diagnostic equipment check for related DTCs and refer to the relevant DTC index - Check the fuel level, add fuel if required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P132B-36	Turbocharger /Supercharger Boost Control A Performance - Signal frequency too low	- Restriction of bypass blade due to debris, fouling or incorrect assembly - Bypass blade actuator failure - Bypass blade failure	- Using the Jaguar Land Rover approved diagnostic equipment check for related DTCs and refer to the relevant DTC index - Check for unrestricted movement of the bypass blade - Check for correct assembly of bypass blade and end stop mechanism - Check and install new bypass blade actuator as required - Check and install new bypass blade as required
P132B-77	Turbocharger /Supercharger Boost Control A Performance	- Restriction of bypass blade due to debris, fouling or incorrect assembly - Bypass blade actuator failure	- Using the Jaguar Land Rover approved diagnostic equipment check for related DTCs and refer to the relevant DTC index - Check for unrestricted movement of the bypass blade

	- Commanded position not reachable	■ Bypass blade failure	■ Check for correct assembly of bypass blade and end stop mechanism ■ Check and install new bypass blade actuator as required ■ Check and install new bypass blade as required
P1338-38	Fuel Pump Driver Module Communication Circuit (Fuel Pump Driver Module) - Signal frequency incorrect	■ The powertrain control module measured an incorrect number of cycles in a given time period ■ Fuel pump driver module circuit, short circuit to ground, short circuit to power, open circuit, high resistance	■ Refer to electrical circuit diagrams and check the fuel pump driver module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the Jaguar Land Rover approved diagnostic equipment check for related DTCs and refer to the relevant DTC index
P1500-31	Vehicle Speed Sensor - No signal	■ Signal error for vehicle speed over CAN	■ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for DTCs and refer to the relevant DTC index
P1592-42	Vehicle Data Recorder Data Available - General memory failure	NOTE: **Monitor description.** Engine control module fails to write flight recorder data to internal memory ■ Battery disconnected during flight recorder data write sequence ■ Power and ground connections to the engine control module, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Engine control module failure	■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check the power and ground connections to the module ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required ■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"

P1592-81	Vehicle Data Recorder Data Available - Invalid serial data received	NOTE: **Monitor description.** Engine control module fails to write flight recorder data to internal memory ■ Battery disconnected during flight recorder data write sequence ■ Power and ground connections to the engine control module, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Engine control module failure	■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check the power and ground connections to the module ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required ■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P162F-00	Starter Motor Disabled - Engine Crank Time Too Long - No sub type information	■ Vehicle battery failure ■ Other starting related failures	■ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ■ Using the Jaguar Land Rover approved diagnostic equipment, check the engine control module, for related DTCs and refer to the relevant DTC index
P1655-04	Starter Disable Circuit - System Internal Failures	■ Engine control module failure	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required

			■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P1655-49	Starter Disable Circuit - Internal electronic failure	■ Engine control module failure	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required ■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P167F-00	Non-OEM Calibration Detected - No sub type information	■ Non-OEM calibration has been flashed into the engine control module	■ No dealer action required - DTC sets when an unauthorised calibration has been flashed into the engine control module
P168F-00	Cold Start Injection System Performance - No sub type information	**NOTE:** **Monitor description.** Split injection during cold start emissions reduction not achieved ■ Other related DTCs ■ Fuel injector failure ■ Engine control module failure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Allow an overnight soak and carry out a cold start, let vehicle idle until catalyst heating has finished (idle speed drops to below 1000rpm) ■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Refer to the electrical circuit diagrams and check fuel injector connections are secure and wiring integrity ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Allow an overnight soak and carry out a cold start. Check engine control module for this DTC ■ Check and install new engine control module as required if the DTC has reset post the cold start

			■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P1801-13	Transmission Clutch Interlock Safety Switch Open Circuit - Circuit open	**NOTE:** Circuit reference - I_S_STREN - ■ Powertrain control module ground circuit open circuit, high resistance	■ Refer to the electrical circuit diagrams and check the powertrain control module ground circuit for open circuit, high resistance. Repair the wiring harness as necessary
P188E-4B	Oil Pressure Pump Control Circuit - Over temperature	■ The powertrain control module detected an internal temperature above the expected range ■ Connector is disconnected, connector terminal is backed out, connector terminal corrosion ■ Variable oil pump circuit, short circuit to ground, short circuit to power, open circuit, high resistance ■ Variable oil pump failure	■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check variable oil pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest ■ Check and install a new variable oil pump as required
P2065-29	Fuel Level Sensor B Circuit - Signal invalid	**NOTE:** **Monitor description.** Open circuit detected by the instrument cluster of the passive fuel level sensor. The instrument cluster will transmit the failure over CAN to the engine control module. The	■ Check connector is not disconnected, connector pin is not backed out ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check fuel level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest ■ Check and install new fuel level sensor as required

		engine control module sets the DTC - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel level sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Fuel level sensor failure	
P2066-00	Fuel Level Sensor B Circuit Range /Performance - No sub type information	NOTE: **Monitor description.** Within the engine control module software a fuel level model is created. If the error between the fuel level model and the measured fuel level has gone over a threshold, and no change in fuel level is detected a DTC is set - The fuel tank passive side fuel level sensor is stuck	- Check the fuel level sensor for correct movement across the entire range
P2088-00	A Camshaft Position Actuator Control Circuit Low Bank 1 - No sub type information	- Variable camshaft timing solenoid circuit short circuit to ground - Variable camshaft timing solenoid failure	- Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new variable camshaft timing solenoid as required

P2089-00	A Camshaft Position Actuator Control Circuit High Bank 1 - No sub type information	- Variable camshaft timing solenoid circuit short circuit to power - Variable camshaft timing solenoid failure	- Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new variable camshaft timing solenoid as required
P2090-00	B Camshaft Position Actuator Control Circuit Low Bank 1 - No sub type information	- Variable camshaft timing solenoid circuit short circuit to ground - Variable camshaft timing solenoid failure	- Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new variable camshaft timing solenoid as required
P2091-00	B Camshaft Position Actuator Control Circuit High Bank 1 - No sub type information	NOTE: **Monitor description.** Exhaust camshaft actuator bank 1 short circuit to power - Variable camshaft timing solenoid circuit short circuit to power - Variable camshaft timing solenoid failure	- Vehicle Conditions to enable DTC Logging strategy - Start and idle engine for at least 2 seconds - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new variable camshaft timing solenoid as required
P2091-85	B Camshaft Position Actuator Control Circuit High Bank 1 - Signal above allowable range	- Variable camshaft timing solenoid circuit short circuit to power - Variable camshaft timing solenoid failure	- Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new variable camshaft timing solenoid as required
P2092-00	A Camshaft Position Actuator Control Circuit	Variable camshaft timing solenoid circuit short circuit to ground	- Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground - Using the Jaguar Land Rover approved diagnostic equipment clear all stored

	Low Bank 2 - No sub type information	■ Variable camshaft timing solenoid failure	DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing solenoid as required
P2093-00	A Camshaft Position Actuator Control Circuit High Bank 2 - No sub type information	■ Variable camshaft timing solenoid circuit short circuit to power ■ Variable camshaft timing solenoid failure	■ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing solenoid as required
P2094-00	B Camshaft Position Actuator Control Circuit Low Bank 2 - No sub type information	■ Variable camshaft timing solenoid circuit short circuit to ground ■ Variable camshaft timing solenoid failure	■ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing solenoid as required
P2095-00	B Camshaft Position Actuator Control Circuit High Bank 2 - No sub type information	NOTE: **Monitor description.** Exhaust camshaft actuator bank 2 short circuit to power ■ Variable camshaft timing solenoid circuit short circuit to power ■ Variable camshaft timing solenoid failure	■ Vehicle Conditions to enable DTC Logging strategy ■ Start and idle engine for at least 2 seconds ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing solenoid as required
P2095-85	B Camshaft Position Actuator Control Circuit High Bank 2 - Signal above allowable range	■ Variable camshaft timing solenoid circuit short circuit to power ■ Variable camshaft timing solenoid failure	■ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to power ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new variable camshaft timing solenoid as required

P2096-00	O2 Sensor Signal Biassed /Stuck Lean - Bank 1, Sensor 1 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained - Exhaust system leakage - Post catalyst heated oxygen sensor incorrectly installed or harness failure - Pre catalyst heated oxygen sensor incorrectly installed or harness failure - Post catalyst oxygen sensor damaged or contaminated internally - Pre catalyst oxygen sensor damaged or contaminated internally - Post catalyst oxygen sensor failure - Pre catalyst oxygen sensor failure	NOTE: Vehicle conditions to enable DTC logging stratagy. Drive vehicle under steady state condition for up to 20 minutes - Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to Electronic Engine Controls V8 S /C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A
P2096-02	O2 Sensor Signal Biassed /Stuck Lean - Bank 1, Sensor 1 - General signal failure	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained - Exhaust system leak - Post catalyst oxygen sensor incorrectly	NOTE: Vehicle conditions to enable DTC logging stratagy. Drive vehicle under steady state condition for up to 20 minutes - Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check Heated Oxygen Sensor

		installed or harness failure ▪ Pre catalyst oxygen sensor incorrectly installed or harness failure ▪ Post catalyst oxygen sensor damaged or contaminated internally ▪ Pre catalyst oxygen sensor damaged or contaminated internally ▪ Post catalyst oxygen sensor failure ▪ Pre catalyst oxygen sensor failure	(HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to Electronic Engine Controls V8 S /C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A
P2097-00	O2 Sensor Signal Biassed /Stuck Rich - Bank 1, Sensor 1 - No sub type information	**NOTE:** **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained ▪ Exhaust system leak ▪ Post catalyst oxygen sensor incorrectly installed or harness failure ▪ Pre catalyst oxygen sensor incorrectly installed or harness failure ▪ Post catalyst oxygen sensor damaged or contaminated internally ▪ Pre catalyst oxygen sensor damaged or contaminated internally ▪ Post catalyst oxygen sensor failure ▪ Pre catalyst oxygen sensor failure	**NOTE:** Vehicle conditions to enable DTC logging stratagy. Drive vehicle under steady state condition for up to 20 minutes ▪ Check the exhaust system for leaks. Repair as necessary ▪ Check pre and post catalyst oxygen sensors for damaged and correct installation ▪ Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to Electronic Engine Controls V8 S /C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A

P2097-02	O2 Sensor Signal Biassed /Stuck Rich - Bank 1, Sensor 1 - General signal failure	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained - Exhaust system leak - Post catalyst oxygen sensor incorrectly installed or harness failure - Pre catalyst oxygen sensor incorrectly installed or harness failure - Post catalyst oxygen sensor damaged or contaminated internally - Pre catalyst oxygen sensor damaged or contaminated internally - Post catalyst oxygen sensor failure - Pre catalyst oxygen sensor failure	NOTE: Vehicle conditions to enable DTC logging stratagy. Drive vehicle under steady state condition for up to 20 minutes - Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to Electronic Engine Controls V8 S /C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A
P2098-00	Post Catalyst Fuel Trim System Too Lean Bank 2 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained - Exhaust system leak	NOTE: Vehicle conditions to enable DTC logging stratagy. Drive vehicle under steady state condition for up to 20 minutes - Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		■ Post catalyst oxygen sensor incorrectly installed or harness failure ■ Pre catalyst oxygen sensor incorrectly installed or harness failure ■ Post catalyst oxygen sensor damaged or contaminated internally ■ Pre catalyst oxygen sensor damaged or contaminated internally ■ Post catalyst oxygen sensor failure ■ Pre catalyst oxygen sensor failure	■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to Electronic Engine Controls V8 S /C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A
P2098-02	O2 Sensor Signal Biassed /Stuck Lean - Bank 2, Sensor 1 - General signal failure	**NOTE:** **Monitor description.** Detects when secondary fuelling adaptation offset is too large to maintain emissions within limits ■ Exhaust system leakage between pre catalyst Heated Oxygen Sensor (HO2S) and mid catalyst Heated Oxygen Sensor (HO2S) ■ Pre catalyst Heated Oxygen Sensor (HO2S) not installed correctly ■ Mid catalyst Heated Oxygen Sensor (HO2S) not installed correctly ■ Mid catalyst Heated Oxygen Sensor (HO2S) installed in incorrect bank ■ Connector is disconnected,	■ Vehicle Conditions to enable DTC Logging strategy ■ Drive vehicle under steady state conditions for up to 20 minutes ■ Prioritised Checks to Perform ■ Check the exhaust system for leaks. Repair as necessary ■ Check pre catalyst Heated Oxygen Sensor (HO2S) for correct installation ■ Check mid catalyst Heated Oxygen Sensor (HO2S) for correct installation ■ Check mid catalyst Heated Oxygen Sensor (HO2S) is installed in correct bank ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Refer to the electrical circuit diagrams and check pre catalyst Heated Oxygen Sensor (HO2S) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Refer to the electrical circuit diagrams and check mid catalyst Heated Oxygen Sensor (HO2S) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Check pre catalyst Heated Oxygen Sensor (HO2S) for damage or contamination ■ Check mid catalyst Heated Oxygen Sensor (HO2S) for damage or contamination ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored

		connector pin is backed out, connector pin corrosion ▪ Pre catalyst Heated Oxygen Sensor (HO2S) circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Mid catalyst Heated Oxygen Sensor (HO2S) circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Pre catalyst Heated Oxygen Sensor (HO2S) damaged or contaminated internally ▪ Mid catalyst Heated Oxygen Sensor (HO2S) damaged or contaminated internally	DTCs using the 'Diagnosis Menu' tab and retest
P2099-00	Post Catalyst Fuel Trim System Too Rich Bank 2 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too small for fuel mixture control to be maintained ▪ Exhaust system leak ▪ Post catalyst oxygen sensor incorrectly installed or harness failure ▪ Pre catalyst oxygen sensor incorrectly installed or harness failure ▪ Post catalyst oxygen sensor damaged or contaminated internally	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle under steady state conditions for up to 20 minutes ▪ Prioritised Checks to Perform ▪ Check the exhaust system for leaks. Repair as necessary ▪ Check pre and post catalyst oxygen sensors for damaged and correct installation ▪ Refer to the electrical circuit diagrams and check post catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ▪ Refer to the electrical circuit diagrams and check pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new pre catalyst oxygen sensor as required ▪ Check and install new post catalyst oxygen sensor as required

		■ Pre catalyst oxygen sensor damaged or contaminated internally ■ Post catalyst oxygen sensor failure ■ Pre catalyst oxygen sensor failure	
P2099-02	O2 Sensor Signal Biassed /Stuck Rich - Bank 2, Sensor 1 - General signal failure	**NOTE:** **Monitor description.** Detects when secondary fuelling adaptation offset is too small to maintain emissions within limits ■ Exhaust system leak ■ Post catalyst oxygen sensor incorrectly installed or harness failure ■ Pre catalyst oxygen sensor incorrectly installed or harness failure ■ Post catalyst oxygen sensor damaged or contaminated internally ■ Pre catalyst oxygen sensor damaged or contaminated internally ■ Post catalyst oxygen sensor failure ■ Pre catalyst oxygen sensor failure	■ Vehicle Conditions to enable DTC Logging strategy ■ Drive vehicle under steady state conditions for up to 20 minutes ■ Prioritised Checks to Perform ■ Check the exhaust system for leaks. Repair as necessary ■ Check pre and post catalyst oxygen sensors for damaged and correct installation ■ Refer to the electrical circuit diagrams and check post catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ■ Refer to the electrical circuit diagrams and check pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new pre catalyst oxygen sensor as required ■ Check and install new post catalyst oxygen sensor as required
P2100-00	Throttle Actuator A Control Motor Circuit /Open - No sub type information	■ Electric throttle actuator control circuit open circuit ■ Electric throttle actuator failure	■ Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for open circuit ■ Refer to the electrical circuit diagrams and check for open circuit across electric throttle power and ground connections with harness disconnected

			- Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new electric throttle actuator as required
P2101-00	Throttle Actuator A Control Motor Circuit Range /Performance - No sub type information	- Electric throttle actuator control power and ground circuits short circuit to each another - Electric throttle actuator failure	- Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other - Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new electric throttle actuator as required
P2102-00	Throttle Actuator A Control Motor Circuit Low - No sub type information	- Electric throttle control circuits short circuit to power, short circuit to ground - Electric throttle actuator control power and ground circuits short circuit to each another - Electric throttle actuator failure	- Refer to the electrical circuit diagrams and check electric throttle actuator circuits for short circuit to power, short circuit to ground - Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other - Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new electric throttle actuator as required
P2103-85	Throttle Actuator A Control Motor Circuit High - Signal above allowable range	- Electric throttle control circuits short circuit to power, short circuit to ground - Electric throttle actuator control power and ground circuits short circuit to each another - Electric throttle actuator failure	- Refer to the electrical circuit diagrams and check electric throttle actuator circuits for short circuit to power, short circuit to ground - Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other - Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected

			▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2108-00	Throttle Actuator, A, Control Module Performance - No sub type information	▪ Electric throttle control circuits short circuit to power, short circuit to ground ▪ Electric throttle actuator control power and ground circuits short circuit to each another ▪ Electric throttle actuator failure	▪ Refer to the electrical circuit diagrams and check electric throttle actuator circuits for short circuit to power, short circuit to ground ▪ Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2111-00	Throttle Actuator "A" Control System - Stuck Open - No sub type information	▪ Electric throttle blocked or restricted ▪ Electric throttle control circuits short circuit to power, short circuit to ground ▪ Electric throttle actuator control power and ground circuits short circuit to each another ▪ Electric throttle actuator failure	▪ Check electric throttle for blockages or obstruction that prevent correct movement ▪ Refer to the electrical circuit diagrams and check electric throttle actuator circuits for short circuit to power, short circuit to ground ▪ Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2112-85	Throttle Actuator "A" Control System - Stuck Closed - Signal above allowable range	▪ Electric throttle blocked or restricted ▪ Electric throttle control circuits short circuit to power, short circuit to ground	▪ Check electric throttle for blockages or obstruction that prevent correct movement ▪ Refer to the electrical circuit diagrams and check electric throttle actuator circuits for short circuit to power, short circuit to ground

		▪ Electric throttle actuator control power and ground circuits short circuit to each another ▪ Electric throttle actuator failure	▪ Refer to the electrical circuit diagrams and check electric throttle actuator power and ground circuits for short circuit to each other ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2119-00	Throttle Actuator "A" Control Throttle Body Range /Performance - No sub type information	▪ Other related DTCs ▪ Electric throttle return spring failure ▪ Electric throttle blocked or restricted ▪ Electric throttle actuator failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check electric throttle return spring for correct operation ▪ Check electric throttle for blockages or obstruction that prevent correct movement ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2119-92	Throttle Actuator "A" Control Throttle Body Range /Performance - Performance or incorrect operation	▪ Other related DTCs ▪ Electric throttle blocked or restricted ▪ Electric throttle actuator failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check electric throttle for blockages or obstruction that prevent correct movement ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required
P2119-97	Throttle Actuator "A" Control Throttle Body Range /Performance - Component or system operation obstructed or blocked	▪ Other related DTCs ▪ Electric throttle blocked or restricted ▪ Electric throttle actuator failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check electric throttle for blockages or obstruction that prevent correct movement ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required

P2122-00	Throttle/Pedal Position Sensor /Switch D Circuit Low - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ This DTC is set when the engine control module detects the pedal demand 1 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new accelerator pedal position sensor as required
P2123-00	Throttle/Pedal Position Sensor /Switch D Circuit High - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ This DTC is set when the engine control module detects the pedal demand 1 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new accelerator pedal position sensor as required
P2127-00	Throttle/Pedal Position Sensor /Switch E Circuit Low - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ This DTC is set when the engine control module detects the pedal demand 2 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new accelerator pedal position sensor as required
P2128-00	Throttle/Pedal Position Sensor /Switch E Circuit High - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ This DTC is set when the engine control module detects the pedal demand 2 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic

			equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new accelerator pedal position sensor as required
P2135-00	Throttle/Pedal Position Sensor /Switch A / B Voltage Correlation - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check and install new accelerator pedal position sensor as required
P2138-00	Throttle/Pedal Position Sensor /Switch D / E Voltage Correlation - No sub type information	▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ This DTC is set when the engine control module detects an excessive difference between the pedal demand signal 1 and signal 2. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensors. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new accelerator pedal position sensor as required
P2146-04	Fuel Injector Group A Supply Voltage Circuit / Open - System Internal Failures	**NOTE:** **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2146-13	Fuel Injector Group A Supply Voltage Circuit Open - Circuit open	▪ Other related DTCs ▪ Connector is disconnected, connector pin is	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		backed out, connector pin corrosion ■ Harness failure - Wiring integrity	■ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2147-04	Fuel Injector Group A Supply Voltage Circuit Low - System Internal Failures	NOTE: **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module ■ Engine control module failure	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required ■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2149-04	Fuel Injector Group B Supply Voltage Circuit / Open - System Internal Failures	NOTE: **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module ■ Engine control module failure	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new engine control module as required ■ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2149-13	Fuel Injector Group B Supply Voltage Circuit Open - Circuit open	■ Other related DTCs ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		- Harness failure - Wiring integrity	- Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2150-04	Fuel Injector Group B Supply Voltage Circuit Low - System Internal Failures	NOTE: **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module - Engine control module failure	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2152-04	Fuel Injector Group C Supply Voltage Circuit / Open - System Internal Failures	NOTE: **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module - Engine control module failure	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2152-13	Fuel Injector Group C Supply Voltage Circuit Open - Circuit open	- Other related DTCs - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		▪ Harness failure - Wiring integrity	▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2155-04	Fuel Injector Group D Supply Voltage Circuit / Open - System Internal Failures	NOTE: **Monitor description.** To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the injector control module ▪ Engine control module failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2155-13	Fuel Injector Group D Supply Voltage Circuit Open - Circuit open	▪ Other related DTCs ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2162-26	Vehicle Speed Sensor A / B Correlation - Signal rate of change below threshold	▪ Vehicle speed comparison difference between transmission control module and anti-lock brake system control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related DTCs and refer to the relevant DTC index

			- Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest
P2162-27	Vehicle Speed Sensor A / B Correlation - Signal rate of change above threshold	Vehicle speed comparison difference between transmission control module and anti-lock brake system control module	- Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest
P2169-13	Exhaust Pressure Regulator Vent Solenoid Control Circuit / Open - Circuit open	- Active exhaust solenoid valve circuit open circuit - Active exhaust solenoid failure	- Refer to the electrical circuit diagrams and check active exhaust solenoid valve circuit for open circuit - Check and install new active exhaust solenoid as required
P2170-11	Exhaust Pressure Regulator Vent Solenoid Control Circuit Low - Circuit short to ground	- Active exhaust solenoid valve circuit short circuit to ground - Active exhaust solenoid failure	- Refer to the electrical circuit diagrams and check active exhaust solenoid valve circuit for short circuit to ground - Check and install new active exhaust solenoid as required
P2170-4B	Exhaust Pressure Regulator Vent Solenoid Control Circuit Low - Over temperature	- Active exhaust solenoid valve circuit short circuit to power, short circuit to ground - Active exhaust solenoid failure	- Refer to the electrical circuit diagrams and check active exhaust solenoid valve circuit for open circuit - Check and install new active exhaust solenoid as required
P2171-12	Exhaust Pressure Regulator Vent Solenoid Control Circuit	Active exhaust solenoid valve circuit short circuit to power	Refer to the electrical circuit diagrams and check active exhaust solenoid valve circuit for short circuit to power

	high - Circuit short to battery	▪ Active exhaust solenoid failure	▪ Check and install new active exhaust solenoid as required
P2176-00	Throttle Actuator "A" Control System - Idle Position Not Learned - No sub type information	▪ Electric throttle actuator adaption routine not completed ▪ Replacement throttle fitted but adaption routine not completed	▪ Allow engine control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2176-51	Throttle Actuator "A" Control System - Idle Position Not Learned - Not programmed	▪ Voltage too low to perform electric throttle actuator adaption routine ▪ Electric throttle actuator adaption routine not completed	▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Allow engine control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2176-52	Throttle Actuator "A" Control System - Idle Position Not Learned - Not activated	▪ Electric throttle actuator adaption routine entry conditions are not met ▪ Electric throttle actuator adaption routine not completed ▪ Replacement throttle fitted but adaption routine not completed	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Allow engine control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2176-54	Throttle Actuator "A" Control System - Idle Position Not Learned - Missing calibration	▪ Electric throttle actuator adaption routine entry conditions are not met ▪ Electric throttle actuator adaption routine not completed ▪ Replacement throttle fitted but adaption routine not completed ▪ Electric throttle actuator failure	▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Allow engine control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new electric throttle actuator as required

P2177-00	System Too Lean Off Idle - Bank 1 - No sub type information	▪ Air intake system leakage ▪ Mass air flow sensor contamination ▪ Exhaust system leakage ▪ Incorrect fuel pressure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position) ▪ Check for air intake system leakage ▪ Check for mass air flow sensor contamination ▪ Check for exhaust system leakage ▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2178-00	System Too Rich Off Idle - Bank 1 - No sub type information	▪ Blocked air filter ▪ Air intake system restriction ▪ Mass air flow sensor contamination ▪ Exhaust system restriction ▪ Incorrect fuel pressure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position) ▪ Check for blocked air filter ▪ Check for air intake system restriction ▪ Check for mass air flow sensor contamination ▪ Check for exhaust system restriction ▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2179-00	System Too Lean Off Idle -	▪ Air intake system leakage	**NOTE:**

	Bank 2 - No sub type information	■ Mass air flow sensor contamination ■ Exhaust system leakage ■ Incorrect fuel pressure	Operational requirements needed to allow the monitor to be fully tested. Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position) ■ Check for air intake system leakage ■ Check for mass air flow sensor contamination ■ Check for exhaust system leakage ■ Refer to the relevant section of the workshop manual and check fuel pressure is to specification ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2180-00	System Too Rich Off Idle - Bank 2 - No sub type information	■ Blocked air filter ■ Air intake system restriction ■ Mass air flow sensor contamination ■ Exhaust system restriction ■ Incorrect fuel pressure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position) ■ Check for blocked air filter ■ Check for air intake system restriction ■ Check for mass air flow sensor contamination ■ Check for exhaust system restriction ■ Refer to the relevant section of the workshop manual and check fuel pressure is to specification ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2183-00	Engine Coolant Temperature Sensor 2 Circuit Range /Performance - No sub type information	NOTE: **Monitor description.** Engine coolant sensor 2 long	■ Prioritised Checks to Perform ■ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Engine Coolant Temperature #2 (0x0489) ■ Record temperatures of coolant 2 watching for movement drive

		term average stuck. No movement in sensor value during drive cycle ▪ Blockage in coolant system ▪ Insufficient mixture of antifreeze creating possibility of icing in coolant system ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Engine coolant temperature sensor failure	vehicle 3 counts fast and 3 counts slow, if no movement in temperature sensor failure ▪ Check coolant system for blockages or restrictions that prevent coolant flow ▪ Check for the correct level of antifreeze within the coolant ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check and install new engine coolant temperature sensor as required
P2183-21	Engine Coolant Temperature Sensor 2 Circuit Range /Performance - Signal amplitude < minimum	▪ Blockage in coolant system ▪ Insufficient mixture of antifreeze creating possibility of icing in coolant system ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Engine coolant temperature sensor failure	▪ Check coolant system for blockages or restrictions that prevent coolant flow ▪ Check for the correct level of antifreeze within the coolant ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check and install new engine coolant temperature sensor as required
P2183-22	Engine Coolant Temperature Sensor 2	▪ Blockage in coolant system	▪ Check coolant system for blockages or restrictions that prevent coolant flow

	Circuit Range /Performance - Signal amplitude > maximum	▪ Insufficient mixture of antifreeze creating possibility of icing in coolant system ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Engine coolant temperature sensor failure	▪ Check for the correct level of antifreeze within the coolant ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check and install new engine coolant temperature sensor as required
P2184-00	Engine Coolant Temperature Sensor 2 Circuit Low - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to ground, open circuit ▪ Engine coolant temperature sensor failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor for short circuit to ground, open circuit ▪ Check and install new engine coolant temperature sensor as required
P2185-00	Engine Coolant Temperature Sensor 2 Circuit High - No sub type information	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, open circuit ▪ Engine coolant temperature sensor failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor for short circuit to power, open circuit ▪ Check and install new engine coolant temperature sensor as required
P2187-00	System Too Lean at Idle - Bank 1 - No sub type information	▪ Air intake system leakage ▪ Pipe detached/union leakage between intake manifold and cylinder head	▪ Check for air intake system leakage ▪ Check intake manifold to cylinder head union for air leaks or pipe disconnected

		- Fuel pressure too low - Mass air flow sensor contaminated - Exhaust system leakage	- Refer to the relevant section of the workshop manual and check fuel pressure is to specification - Check for mass air flow sensor contamination - Check for exhaust system leakage - Check all heated oxygen sensors are installed correctly - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2188-00	System Too Rich at Idle - Bank 1 - No sub type information	- Air filter blocked - Air intake system restriction - Fuel pressure too high - Mass air flow sensor contaminated - Exhaust system blocked	- Check air filter is not blocked - Check the intake air system is free from restrictions - Refer to the relevant section of the workshop manual and check fuel pressure is to specification - Check for mass air flow sensor contamination - Check for exhaust system blockage - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2189-00	System Too Lean at Idle - Bank 2 - No sub type information	- Air intake system leakage - Pipe detached/union leakage between intake manifold and cylinder head - Fuel pressure too low - Mass air flow sensor contaminated - Exhaust system leakage	- Check for air intake system leakage - Check intake manifold to cylinder head union for air leaks or pipe disconnected - Refer to the relevant section of the workshop manual and check fuel pressure is to specification - Check for mass air flow sensor contamination - Check for exhaust system leakage - Check all heated oxygen sensors are installed correctly - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2190-00	System Too Rich at Idle - Bank 2 - No sub type information	- Air filter blocked - Air intake system restriction - Fuel pressure too high - Mass air flow sensor contaminated	- Check air filter is not blocked - Check the intake air system is free from restrictions - Refer to the relevant section of the workshop manual and check fuel pressure is to specification - Check for mass air flow sensor contamination

		▪ Exhaust system blocked	▪ Check for exhaust system blockage ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2195-00	O2 Sensor Signal Biassed /Stuck Lean - Bank 1, Sensor 1 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained ▪ Exhaust system leakage ▪ Post catalyst heated oxygen sensor incorrectly installed ▪ Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Pre catalyst heated oxygen sensor contamination or failure ▪ Post catalyst heated oxygen sensor contamination or failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes ▪ Check exhaust system for leakage. Rectify as required ▪ Check post catalyst heated oxygen sensor is correctly installed ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new pre catalyst heated oxygen sensor as required ▪ Check and install new post catalyst heated oxygen sensor as required
P2196-00	O2 Sensor Signal Biassed /Stuck Rich - Bank 1, Sensor 1 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation	NOTE: Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes

		offset is too small for fuel mixture control to be maintained - Exhaust system leakage - Post catalyst heated oxygen sensor incorrectly installed - Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Pre catalyst heated oxygen sensor contamination or failure - Post catalyst heated oxygen sensor contamination or failure	- Check exhaust system for leakage. Rectify as required - Check post catalyst heated oxygen sensor is correctly installed - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new pre catalyst heated oxygen sensor as required - Check and install new post catalyst heated oxygen sensor as required
P2197-00	O2 Sensor Signal Biassed /Stuck Lean - Bank 2, Sensor 1 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too large for fuel mixture control to be maintained - Exhaust system leak - Post catalyst oxygen sensor incorrectly installed or harness failure	- Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check post catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance - Refer to the electrical circuit diagrams and check pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new pre catalyst oxygen sensor as required - Check and install new post catalyst oxygen sensor as required

		- Pre catalyst oxygen sensor incorrectly installed or harness failure - Post catalyst oxygen sensor damaged or contaminated internally - Post catalyst oxygen sensor failure - Pre catalyst oxygen sensor failure	
P2198-00	O2 Sensor Signal Biassed /Stuck Rich - Bank 2, Sensor 1 - No sub type information	NOTE: **Monitor description.** Detects when secondary fuelling adaptation offset is too small for fuel mixture control to be maintained - Exhaust system leak - Post catalyst oxygen sensor incorrectly installed or harness failure - Pre catalyst oxygen sensor incorrectly installed or harness failure - Post catalyst oxygen sensor damaged or contaminated internally - Post catalyst oxygen sensor failure - Pre catalyst oxygen sensor failure	- Check the exhaust system for leaks. Repair as necessary - Check pre and post catalyst oxygen sensors for damaged and correct installation - Refer to the electrical circuit diagrams and check post catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance - Refer to the electrical circuit diagrams and check pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new pre catalyst oxygen sensor as required - Check and install new post catalyst oxygen sensor as required
P219A-84	Bank 1 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50°C, drive vehicle at steady load in 6th gear to achieve

		threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Intake air system blockage, restricting air from entering cylinders - Specified cylinder, localised damage /restriction of fuel rail - Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders - Check for damage/restriction of fuel rail - Check for blockage/restriction within exhaust manifold - Check and install new fuel injector as required - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219A-85	Bank 1 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50°C, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first

		the overall fuel system monitor ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Specified cylinder, localised air leak on intake manifold to cylinder head connection ▪ External fuel leak at fuel injector ▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check for air leak on intake manifold to cylinder head connection ▪ Check for external fuel leak at fuel injector ▪ Check for fuel restriction within fuel injector ▪ Check for damage/restriction of fuel rail ▪ Check for air leak around spark plug ▪ Check for air leak around fuel injector ▪ Check and install new fuel injector as required ▪ Check and install new ignition coil as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219B-84	Bank 2 Air-Fuel Ratio Imbalance - Signal below allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50°C, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first

		the overall fuel system monitor ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ■ Intake air system blockage, restricting air from entering cylinders ■ Specified cylinder, localised damage /restriction of fuel rail ■ Specified cylinder, localised blockage /restriction within exhaust manifold ■ Fuel injector failure ■ Damage/friction in valve train components ■ Camshaft profile switching system failure	■ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ■ Check for damage/restriction of fuel rail ■ Check for blockage/restriction within exhaust manifold ■ Check and install new fuel injector as required ■ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
P219B-85	Bank 2 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ■ The engine control module has determined failures	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50°C, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ■ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check for air leak on intake manifold to cylinder head connection ■ Check for external fuel leak at fuel injector

		where some circuit quantity, reported via serial data, is above a specified range ▪ Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Specified cylinder, localised air leak on intake manifold to cylinder head connection ▪ External fuel leak at fuel injector ▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	▪ Check for fuel restriction within fuel injector ▪ Check for damage/restriction of fuel rail ▪ Check for air leak around spark plug ▪ Check for air leak around fuel injector ▪ Check and install new fuel injector as required ▪ Check and install new ignition coil as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219C-84	Cylinder 1 Air-Fuel Ratio Imbalance - Signal below allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ▪ Check for damage/restriction of fuel rail ▪ Check for blockage/restriction within exhaust manifold

			- Intake air system blockage, restricting air from entering cylinders - Specified cylinder, localised damage /restriction of fuel rail - Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	- Check and install new fuel injector as required - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
	P219C-85	Cylinder 1 Air-Fuel Ratio Imbalance - Signal above allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on intake manifold to cylinder head connection - External fuel leak at fuel injector	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug - Check for air leak around fuel injector - Check and install new fuel injector as required - Check and install new ignition coil as required

		▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219D-84	Cylinder 2 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ▪ Intake air system blockage, restricting air from entering cylinders ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Specified cylinder, localised blockage /restriction within exhaust manifold ▪ Fuel injector failure ▪ Damage/friction in valve train components ▪ Camshaft profile switching system failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ▪ Check for damage/restriction of fuel rail ▪ Check for blockage/restriction within exhaust manifold ▪ Check and install new fuel injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P219D-85	Cylinder 2 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on intake manifold to cylinder head connection - External fuel leak at fuel injector - Fuel restriction within fuel injector - Specified cylinder, localised damage /restriction of fuel rail - Air leak around spark plug - Air leak around fuel injector - Fuel injector failure - Ignition coil failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug - Check for air leak around fuel injector - Check and install new fuel injector as required - Check and install new ignition coil as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219E-84	Cylinder 3 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve

			(lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Intake air system blockage, restricting air from entering cylinders - Specified cylinder, localised damage /restriction of fuel rail - Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders - Check for damage/restriction of fuel rail - Check for blockage/restriction within exhaust manifold - Check and install new fuel injector as required - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
	P219E-85	Cylinder 3 Air-Fuel Ratio Imbalance - Signal above allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection

		▪ Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Specified cylinder, localised air leak on intake manifold to cylinder head connection ▪ External fuel leak at fuel injector ▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	▪ Check for external fuel leak at fuel injector ▪ Check for fuel restriction within fuel injector ▪ Check for damage/restriction of fuel rail ▪ Check for air leak around spark plug ▪ Check for air leak around fuel injector ▪ Check and install new fuel injector as required ▪ Check and install new ignition coil as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219F-84	Cylinder 4 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ▪ Intake air system blockage, restricting air from entering cylinders ▪ Specified cylinder, localised damage /restriction of fuel rail	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ▪ Check for damage/restriction of fuel rail ▪ Check for blockage/restriction within exhaust manifold ▪ Check and install new fuel injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine

		- Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P219F-85	Cylinder 4 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on intake manifold to cylinder head connection - External fuel leak at fuel injector - Fuel restriction within fuel injector - Specified cylinder, localised damage /restriction of fuel rail - Air leak around spark plug	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug - Check for air leak around fuel injector - Check and install new fuel injector as required - Check and install new ignition coil as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	
P21A0-84	Cylinder 5 Air-Fuel Ratio Imbalance - Signal below allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ▪ Intake air system blockage, restricting air from entering cylinders ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Specified cylinder, localised blockage /restriction within exhaust manifold ▪ Fuel injector failure ▪ Damage/friction in valve train components ▪ Camshaft profile switching system failure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ▪ Check for damage/restriction of fuel rail ▪ Check for blockage/restriction within exhaust manifold ▪ Check and install new fuel injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21A0-85	Cylinder 5 Air-Fuel Ratio Imbalance - Signal above allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda)	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm,

		above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on intake manifold to cylinder head connection - External fuel leak at fuel injector - Fuel restriction within fuel injector - Specified cylinder, localised damage /restriction of fuel rail - Air leak around spark plug - Air leak around fuel injector - Fuel injector failure - Ignition coil failure	maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug - Check for air leak around fuel injector - Check and install new fuel injector as required - Check and install new ignition coil as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21A1-84	Cylinder 6 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine

		which may not be detected by the overall fuel system monitor - Intake air system blockage, restricting air from entering cylinders - Specified cylinder, localised damage /restriction of fuel rail - Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	control module for related misfire DTCs and refer to this DTC index. Rectify these first - Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders - Check for damage/restriction of fuel rail - Check for blockage/restriction within exhaust manifold - Check and install new fuel injector as required - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21A1-85	Cylinder 6 Air-Fuel Ratio Imbalance - Signal above allowable range	**NOTE:** **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug

		intake manifold to cylinder head connection ▪ External fuel leak at fuel injector ▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	▪ Check for air leak around fuel injector ▪ Check and install new fuel injector as required ▪ Check and install new ignition coil as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21A2-84	Cylinder 7 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ▪ Intake air system blockage, restricting air from entering cylinders ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Specified cylinder, localised blockage /restriction within exhaust manifold ▪ Fuel injector failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders ▪ Check for damage/restriction of fuel rail ▪ Check for blockage/restriction within exhaust manifold ▪ Check and install new fuel injector as required ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Damage/friction in valve train components ▪ Camshaft profile switching system failure	
P21A2-85	Cylinder 7 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor ▪ Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Specified cylinder, localised air leak on intake manifold to cylinder head connection ▪ External fuel leak at fuel injector ▪ Fuel restriction within fuel injector ▪ Specified cylinder, localised damage /restriction of fuel rail ▪ Air leak around spark plug ▪ Air leak around fuel injector ▪ Fuel injector failure ▪ Ignition coil failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check for air leak on intake manifold to cylinder head connection ▪ Check for external fuel leak at fuel injector ▪ Check for fuel restriction within fuel injector ▪ Check for damage/restriction of fuel rail ▪ Check for air leak around spark plug ▪ Check for air leak around fuel injector ▪ Check and install new fuel injector as required ▪ Check and install new ignition coil as required ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P21A3-84	Cylinder 8 Air-Fuel Ratio Imbalance - Signal below allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) below threshold, (richer mixture). Monitors cylinder specific fuelling errors which may not be detected by the overall fuel system monitor - Intake air system blockage, restricting air from entering cylinders - Specified cylinder, localised damage /restriction of fuel rail - Specified cylinder, localised blockage /restriction within exhaust manifold - Fuel injector failure - Damage/friction in valve train components - Camshaft profile switching system failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first - Check condition of airpath from throttle to cylinder. Check for intake air system blockage, restricting air from entering cylinders - Check for damage/restriction of fuel rail - Check for blockage/restriction within exhaust manifold - Check and install new fuel injector as required - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related camshaft profile switching system DTCs and refer to this DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21A3-85	Cylinder 8 Air-Fuel Ratio Imbalance - Signal above allowable range	NOTE: **Monitor description.** Inferred mixture strength (lambda) above threshold, (leaner mixture). Monitors cylinder specific fuelling errors	NOTE: Operational requirements needed to allow the monitor to be fully tested. Engine coolant temperature greater than 50 ºC, drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise) - Using the Jaguar Land Rover approved diagnostic equipment check engine control module for related misfire DTCs

		which may not be detected by the overall fuel system monitor - Fuel injector connector is disconnected, connector pin is backed out, connector pin corrosion - Specified cylinder, localised air leak on intake manifold to cylinder head connection - External fuel leak at fuel injector - Fuel restriction within fuel injector - Specified cylinder, localised damage /restriction of fuel rail - Air leak around spark plug - Air leak around fuel injector - Fuel injector failure - Ignition coil failure	and refer to this DTC index. Rectify these first - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check for air leak on intake manifold to cylinder head connection - Check for external fuel leak at fuel injector - Check for fuel restriction within fuel injector - Check for damage/restriction of fuel rail - Check for air leak around spark plug - Check for air leak around fuel injector - Check and install new fuel injector as required - Check and install new ignition coil as required - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2226-16	Barometric Pressure Sensor A Circuit - Circuit voltage below threshold	**NOTE:** The sensor is installed within the powertrain control module and is not serviceable - Powertrain control module failure	- Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P2226-17	Barometric Pressure Sensor A Circuit - Circuit voltage above threshold	**NOTE:** The sensor is installed within the powertrain	- Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

		control module and is not serviceable ▪ Powertrain control module failure	
P2227-00	Barometric Pressure Sensor A Circuit Range /Performance - No sub type information	**NOTE:** The sensor is installed within the engine control module and is not serviceable ▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2227-21	Barometric Pressure Sensor A Circuit Range /Performance - Signal amplitude < minimum	**NOTE:** The sensor is installed within the engine control module and is not serviceable ▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2227-22	Barometric Pressure Sensor A Circuit Range /Performance - Signal amplitude > maximum	**NOTE:** The sensor is installed within the engine control module and is not serviceable ▪ Engine control module failure	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2228-00	Barometric Pressure Sensor A Circuit Low - No sub type information	**NOTE:** The sensor is installed within the engine	▪ Check and install new engine control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"

		control module and is not serviceable - Engine control module failure	
P2229-00	Barometric Pressure Sensor A Circuit High - No sub type information	**NOTE:** The sensor is installed within the engine control module and is not serviceable - Engine control module failure	- Check and install new engine control module as required - Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"
P2231-00	O2 Sensor Signal Circuit Shorted to Heater Circuit - Bank 1, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2234-00	O2 Sensor Signal Circuit Shorted to Heater Circuit - Bank 2, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2237-00	O2 Sensor Positive Current Control Circuit Open - Bank 1, Sensor	**NOTE:**	- Vehicle Conditions to enable DTC Logging strategy - Drive the vehicle in a transient manor, including full load acceleration and overruns to ensure

	1 - No sub type information	**Monitor description.** Detects if the heated oxygen sensor is active during closed loop fuelling control ■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Heated oxygen sensor	the oxygen sensor controller is fully exercised ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required
P2237-02	O2 Sensor Positive Current Control Circuit Open - Bank 1, Sensor 1 - Signal above allowable range	NOTE: **Monitor description.** Detects an open circuit of the heated oxygen sensor output in closed loop fuelling control ■ Heated oxygen sensor circuit open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Heated oxygen sensor	■ Vehicle Conditions to enable DTC Logging strategy ■ Drive vehicle under steady state conditions ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required
P2237-13	O2 Sensor Positive Current Control Circuit / Open - Bank 1, Sensor 1 - Circuit open	■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		- Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor	- Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2237-64	O2 Sensor Positive Current Control Circuit / Open - Bank 1, Sensor 1 - Signal plausibility failure	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2237-85	O2 Sensor Positive Current Control Circuit / Open - Bank 1, Sensor 1 - Signal above allowable range	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2240-00	O2 Sensor Positive Current Control Circuit / Open - Bank 2, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2240-19	O2 Sensor Positive Current Control	- Heated oxygen sensor circuit short circuit to ground,	

	Circuit / Open - Bank 2, Sensor 1 - Circuit current above threshold	short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Heated oxygen sensor failure	■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required
P2240-64	O2 Sensor Positive Current Control Circuit / Open - Bank 2, Sensor 1 - Signal plausibility failure	■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Heated oxygen sensor failure	■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required
P2243-00	O2 Sensor Reference Voltage Circuit / Open - Bank 1, Sensor 1 - No sub type information	■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Heated oxygen sensor failure	■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required
P2247-00	O2 Sensor Reference Voltage Circuit / Open - Bank 2, Sensor 1 - No sub type information	■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ Check and install new heated oxygen sensor as required

		Heated oxygen sensor failure	
P2251-00	O2 Sensor Negative Current Control Circuit / Open - Bank 1, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2254-00	O2 Sensor Negative Current Control Circuit / Open - Bank 2, Sensor 1 - No sub type information	- Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new heated oxygen sensor as required
P2270-00	O2 Sensor Signal Stuck Lean - Bank 1, Sensor 2 - No sub type information	- Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance - Air leakage between heated oxygen sensor and catalyst - Heated oxygen sensor tip damaged - Heated oxygen sensor tip blocked - Heated oxygen sensor tip contaminated by excessive oil consumption - Heated oxygen sensor tip contaminated by poor or incorrect fuel	- Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 2 - (0x035F) - Heated oxygen sensor signal for correct operation - Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check heated oxygen sensor is correctly installed - Check heated oxygen sensor for tip damage - Check heated oxygen sensor for tip blocked - Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check

			▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2271-00	O2 Sensor Signal Stuck Rich - Bank 1, Sensor 2 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 2 - (0x035F) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2272-00	O2 Sensor Signal Stuck Lean - Bank 2, Sensor 2 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 2 - (0x0362) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel

P2273-00	O2 Sensor Signal Stuck Rich - Bank 2, Sensor 2 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 2 - (0x0362) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2274-00	O2 Sensor Signal Stuck Lean - Bank 1, Sensor 3 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 3 - (0x0360) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2275-00	O2 Sensor Signal Stuck Rich - Bank 1, Sensor 3 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 3 - (0x0360) ▪ Heated oxygen sensor signal for correct operation

		▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2276-00	O2 Sensor Signal Stuck Lean - Bank 2, Sensor 3 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 3 - (0x0363) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2277-00	O2 Sensor Signal Stuck Rich - Bank 2, Sensor 3 - No sub type information	▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged	▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 3 - (0x0363) ▪ Heated oxygen sensor signal for correct operation ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed

		▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P2278-00	Oxygen Sensor Signals Swapped Bank 1 Sensor 3 / Bank 2 Sensor 3 - No sub type information	▪ Post catalytic converter heated oxygen sensor connectors swapped bank to bank	▪ Check post catalytic converter heated oxygen sensor connectors are not swapped bank to bank
P2281-21	Air Leak Between MAF and Throttle Body - Signal amplitude < minimum	NOTE: **Monitor description.** Air leak between mass air flow sensor and electric throttle ▪ Air intake system leakage between the mass air flow sensor and the electric throttle	▪ Check for air intake system leakage between the mass air flow sensor and the electric throttle, left bank and right bank
P2281-22	Air Leak Between MAF and Throttle Body - Signal amplitude > maximum	NOTE: **Monitor description.** Air leak between mass air flow sensor and electric throttle ▪ Air intake system blockage between the mass air flow sensor and the electric throttle	▪ Check for air intake system blockage between the mass air flow sensor and the electric throttle, left bank and right bank

P2282-00	Air Leak Between Throttle Body and Intake Valve - No sub type information	▪ Air intake system leakage between the electric throttle and intake valve	▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Check for air intake system leakage between the electric throttle and the intake valve left bank and right bank
P228E-84	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low - Signal below allowable range	▪ Damaged/worn high pressure fuel pump ▪ Fuel pump incorrect assembly/timing ▪ High pressure fuel pump failure	▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Check for a damaged or worn high pressure fuel pump ▪ Check timing and condition of the fuel pump drive chain ▪ Check and install new high pressure fuel pump as required
P228F-85	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High - Signal above allowable range	▪ Damaged/worn high pressure fuel pump ▪ Fuel pump incorrect assembly/timing ▪ High pressure fuel pump failure	▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Check for damaged or worn high pressure fuel pump ▪ Check timing and condition of the fuel pump drive chain ▪ Check and install new high pressure fuel pump as required
P2292-92	Injector Control Pressure Erratic - Performance or incorrect operation	NOTE: **Monitor description.** Loss in high pressure fuel. This can cause problem with stop/start system ▪ Fuel system leakage from the fuel tank and lines ▪ Fuel system leakage into the low fuel pressure system	▪ Check for fuel system leakage from the fuel tank and lines ▪ Check for fuel system leakage from the high pressure fuel system into the low pressures fuel system
P2299-23	Brake Pedal Position /Accelerator Pedal Position Incompatible - Signal stuck low	NOTE: **Monitor description.** If the driver is braking and producing a high brake line pressure whilst	▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Check that the brake pedal switch is mounted and correctly positioned on the mounting bracket/pedal box ▪ Check for smooth operation of the brake pedal switch plunger ▪ Refer to the electrical circuit diagrams and check brake pedal switch 1 circuit

		not pressing the accelerator pedal then both brake pedal switches should be on /active. If the brake pedal switches are off /inactive in this condition then the switches have failed and the DTC is set - Brake pedal switch detached from mounting bracket /pedal box whilst electrically connected - Brake pedal switch mounting position incorrectly adjusted - Brake pedal switch plunger partially stuck in - Brake pedal switch 1 circuit open circuit, short circuit to power, short circuit to ground - Brake pedal switch 2 circuit open circuit, short circuit to power, short circuit to ground	for open circuit, short circuit to power, short circuit to ground - Refer to the electrical circuit diagrams and check brake pedal switch 2 circuit for open circuit, short circuit to power, short circuit to ground
P2299-24	Brake Pedal Position /Accelerator Pedal Position Incompatible - Signal stuck high	NOTE: **Monitor description.** If the driver is pressing the accelerator pedal and the vehicle is accelerating and the brake pressure is low both brake pedal switches should be off /inactive. If the switches are on/active in this condition then the brake pedal switches	- Check engine control module for related DTCs and refer to relevant DTC index - Check that the brake pedal switch is mounted and correctly positioned on the mounting bracket/pedal box - Check for smooth operation of the brake pedal switch plunger - Refer to the electrical circuit diagrams and check brake pedal switch 1 circuit for open circuit, short circuit to power, short circuit to ground - Refer to the electrical circuit diagrams and check brake pedal switch 2 circuit for open circuit, short circuit to power, short circuit to ground

		have failed and the DTC is set - Brake pedal switch detached from mounting bracket /pedal box whilst electrically connected - Brake pedal switch mounting position incorrectly adjusted - Brake pedal switch plunger partially stuck out - Brake pedal switch 1 circuit open circuit, short circuit to power, short circuit to ground - Brake pedal switch 2 circuit open circuit, short circuit to power, short circuit to ground	
P2300-11	Ignition Coil A Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC - Ignition coil circuit, short circuit to ground	- Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground
P2301-12	Ignition Coil A Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module,	- Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power

		passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to power	
P2303-11	Ignition Coil B Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground
P2304-12	Ignition Coil B Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to power	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power
P2306-11	Ignition Coil C Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.**	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground

		Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	
P2307-12	Ignition Coil C Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to power	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power
P2309-11	Ignition Coil D Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground

P2310-12	Ignition Coil D Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC - Ignition coil circuit, short circuit to power	- Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power
P2312-11	Ignition Coil E Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC - Ignition coil circuit, short circuit to ground	- Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground
P2313-12	Ignition Coil E Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC	- Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power

		▪ Ignition coil circuit, short circuit to power	
P2315-11	Ignition Coil F Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground
P2316-12	Ignition Coil F Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to power	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power
P2318-11	Ignition Coil G Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground

		control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	
P2319-12	Ignition Coil G Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to power	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power
P2321-11	Ignition Coil H Primary Control Circuit Low - Circuit short to ground	NOTE: **Monitor description.** Short circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC ▪ Ignition coil circuit, short circuit to ground	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to ground
P2322-12	Ignition Coil H Primary Control Circuit High - Circuit short to battery	NOTE: **Monitor description.** Short circuit detected by	▪ Refer to the electrical circuit diagrams and check ignition coil circuit for short circuit to power

		the ignition control module, passed to main CPU within engine control module and sets the DTC - Ignition coil circuit, short circuit to power	
P2400-00	Evaporative Emission System Leak Detection Pump Control Circuit/Open - No sub type information	- Diagnostic module tank leakage pump circuit open circuit, high resistance	**NOTES:** - If purge valve related DTCs are also set, perform the relevant corrective action(s) first. - It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. - Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for open circuit, high resistance
P2401-00	Evaporative Emission System Leak Detection Pump Control Circuit Low - No sub type information	- Diagnostic module tank leakage pump circuit short circuit to ground	**NOTES:** - If purge valve related DTCs are also set, perform the relevant corrective action(s) first. - It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs.

			▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to ground
P2402-00	Evaporative Emission System Leak Detection Pump Control Circuit High - No sub type information	▪ Diagnostic module tank leakage pump circuit short circuit to power	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to power
P2402-85	Evaporative Emission System Leak Detection Pump Control Circuit High - Signal above allowable range	▪ Diagnostic module tank leakage pump circuit short circuit to power	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to power
P2404-00	Evaporative Emission System Leak Detection Pump Sense	▪ Diagnostic module tank leakage module internal failure ▪ Changeover valve fault	**NOTES:**

	Circuit Range /Performance - No sub type information		■ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ■ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ■ Install a new diagnostic module tank leakage module as necessary
P2405-00	Evaporative Emission System Leak Detection Pump Sense Circuit Low - No sub type information	■ Diagnostic module tank leakage module internal failure	**NOTES:** ■ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ■ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ■ Install a new diagnostic module tank leakage module as necessary
P2406-00	Evaporative Emission System Leak Detection Pump Sense Circuit High - No sub type information	■ Diagnostic module tank leakage module internal failure	**NOTES:** ■ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ■ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs.

			▪ Install a new diagnostic module tank leakage module as necessary
P2407-00	Evaporative Emission System Leak Detection Pump Sense Circuit Intermittent /Erratic - No sub type information	▪ Diagnostic module tank leakage pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diagnostic module tank leakage module internal failure	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new diagnostic module tank leakage module as necessary
P240A-00	Evaporative Emission System Leak Detection Pump Heater Circuit/Open - No sub type information	▪ Diagnostic module tank leakage heater circuit open circuit, high resistance	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for open circuit, high resistance
P240B-00	Evaporative Emission System Leak Detection	▪ Diagnostic module tank leakage heater circuit short circuit to ground	**NOTES:**

	Pump Heater Circuit Low - No sub type information		▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for short circuit to ground
P240C-00	Evaporative Emission System Leak Detection Pump Heater Circuit High - No sub type information	▪ Diagnostic module tank leakage heater circuit short circuit to power	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for short circuit to power
P240C-85	Evaporative Emission System Leak Detection Pump Heater Circuit High - Signal above allowable range	▪ Diagnostic module tank leakage heater circuit short circuit to power	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System

			Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for short circuit to power
P2415-00	O2 Sensor Exhaust Sample Error Bank 2, Sensor 1 - No sub type information	▪ Heated oxygen sensor incorrectly installed or missing ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Check heated oxygen sensor is correctly installed ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new heated oxygen sensor as required
P2418-00	Evaporative Emission Control System Switching Valve Control Circuit/Open - No sub type information	▪ Diagnostic module tank leakage changeover valve circuit open circuit, high resistance	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for open circuit, high resistance
P2419-00	Evaporative Emission Control System Switching Valve Control	▪ Diagnostic module tank leakage changeover valve circuit short circuit to ground	**NOTES:** ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first.

	Circuit Low - No sub type information		▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for short circuit to ground
P2420-00	Evaporative Emission Control System Switching Valve Control Circuit High - No sub type information	▪ Diagnostic module tank leakage changeover valve circuit short circuit to power	NOTES: ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs. ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for short circuit to power
P2420-85	Evaporative Emission Control System Switching Valve Control Circuit High - Signal above allowable range	▪ Diagnostic module tank leakage changeover valve circuit short circuit to power	NOTES: ▪ If purge valve related DTCs are also set, perform the relevant corrective action(s) first. ▪ It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read DTCs.

			- Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for short circuit to power
P250A-12	Engine Oil Level Sensor Circuit - Circuit short to battery	- Oil level sensor circuit short circuit to power	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P250A-23	Engine Oil Level Sensor Circuit - Signal stuck low	NOTE: **Monitor description.** Engine control module monitors the input pin from the oil level sensor and sets the DTC if the input is low - Oil level sensor circuit short circuit to ground, high resistance	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P250A-24	Engine Oil Level Sensor Circuit - Signal stuck high	NOTE: **Monitor description.** Engine control module monitors the input pin from the oil level sensor and sets the DTC if the input is high - Oil level sensor circuit short circuit to power	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P250A-25	Engine Oil Level Sensor Circuit - Signal shape / waveform failure	NOTE: **Monitor description.** Engine control module monitors the PWM signal from the sensor and sets the DTC if an out of range signal is detected - Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Oil level sensor failure	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new oil level sensor as required
P250A-41	Engine Oil Level Sensor Circuit - General checksum failure	NOTE: **Monitor description.** Engine control module monitors the PWM signal from the sensor and sets the DTC if a checksum or watchdog error is detected - Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Oil level sensor failure	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new oil level sensor as required
P250B-00	Engine Oil Level Sensor Circuit Range /Performance - No sub type information	NOTE: **Monitor description.** Engine control module monitors the	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		oil temperature from the sensor and sets the DTC if the temperature is not plausible - Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Oil level sensor failure	- Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new oil level sensor as required
P250B-25	Engine Oil Level Sensor Circuit Range /Performance - Signal shape /waveform failure	NOTE: **Monitor description.** Engine control module monitors the PWM signal from the sensor and sets the DTC if an out of range signal is detected - Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Oil level sensor failure	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new oil level sensor as required
P250B-41	Engine Oil Level Sensor Circuit Range /Performance - General checksum failure	NOTE: **Monitor description.** Engine control module monitors the PWM signal from the sensor and sets the DTC if a checksum or watchdog error is detected	- Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new oil level sensor as required

		■ Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Oil level sensor failure	
P250C-23	Engine Oil Level Sensor Circuit Low - Signal stuck low	NOTE: **Monitor description.** Engine control module monitors the input pin from the oil level sensor and sets the DTC if the input is low ■ The engine control module measures a signal that remains low when transitions are expected ■ Oil level sensor circuit short circuit to ground, high resistance	■ Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P250D-24	Engine Oil Level Sensor Circuit High - Signal stuck high	NOTE: **Monitor description.** Engine control module monitors the input pin from the oil level sensor and sets the DTC if the input is high ■ The engine control module measures a signal that remains high when transitions are expected	■ Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to power ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Oil level sensor circuit short circuit to power	
P252B-00	Engine Oil Quality Sensor Circuit Range /Performance - No sub type information	▪ Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil level sensor failure	▪ Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new oil level sensor as required
P252B-62	Engine Oil Quality Sensor Circuit Range /Performance - Signal compare failure	▪ Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil level sensor failure	▪ Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new oil level sensor as required
P252F-00	Engine Oil Level Too High - No sub type information	NOTE: **Monitor description.** The engine control module sets the DTC to request an oil warm up, due to fuel amount in the oil, over the limit. This DTC will only relate to a flex fuel vehicle (a vehicle that can use up to E85 fuel blend) ▪ The vehicle is displaying the	▪ The engine needs to get to full operating temperature to burn off the fuel, carry out a drive cycle that enables the engine to reach normal operating temperature. To remove the "engine oil level high" message the engine must be fully warmed up and run in a fully warm state for a further 30 minutes ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		"engine oil level high" message - Multiple consecutive short journeys (less than 30 minutes driving) without allowing the engine oil to fully warm up to normal operating temperature	
P2541-00	Low Pressure Fuel System Sensor Circuit Low - No sub type information	NOTE: **Monitor description.** Fuel low pressure sensor voltage below threshold. A break in the feedback of the fuel low pressure control has been detected - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel low pressure sensor circuit, short circuit to ground - Fuel low pressure sensor 5 volt power supply circuit, open circuit, high resistance - Fuel low pressure sensor failure	- Vehicle Conditions to enable DTC Logging strategy - Ignition on engine running and idling for greater than 30 seconds - Prioritised Checks to Perform - Check connector is not disconnected, connector pin is not backed out - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground - Check fuel low pressure sensor 5 volt power supply circuit for open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new fuel low pressure sensor as required
P2541-84	Low Pressure Fuel System Sensor Circuit Low - Signal below allowable range	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel low pressure sensor circuit, short circuit to ground - Fuel low pressure sensor 5 volt power	- Check connector is not disconnected, connector pin is not backed out - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground - Check fuel low pressure sensor 5 volt power supply circuit for open circuit, high resistance

		supply circuit, open circuit, high resistance ▪ Fuel low pressure sensor failure	▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new fuel low pressure sensor as required
P2542-00	Low Pressure Fuel System Sensor Circuit High - No sub type information	NOTE: **Monitor description.** Fuel low pressure sensor voltage above threshold. A break in the feedback of the fuel low pressure control has been detected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel low pressure sensor circuit, short circuit to power ▪ Fuel low pressure sensor ground supply circuit, open circuit, high resistance ▪ Fuel low pressure sensor failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on engine running and idling for greater than 30 seconds ▪ Prioritised Checks to Perform ▪ Check connector is not disconnected, connector pin is not backed out ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to power ▪ Check fuel low pressure sensor ground supply circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new fuel low pressure sensor as required
P2542-85	Low Pressure Fuel System Sensor Circuit High - Signal above allowable range	▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel low pressure sensor circuit, short circuit to power ▪ Fuel low pressure sensor ground supply circuit, open circuit, high resistance ▪ Fuel low pressure sensor failure	▪ Check connector is not disconnected, connector pin is not backed out ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to power ▪ Check fuel low pressure sensor ground supply circuit for open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

			- Check and install new fuel low pressure sensor as required
P2564-00	Turbocharger Boost Control Position Sensor A Circuit Low - No sub type information	- Supercharger bypass valve actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Supercharger bypass valve actuator failure	- Refer to the electrical circuit diagrams and check supercharger bypass valve actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new supercharger bypass valve actuator as required
P2565-00	Turbocharger Boost Control Position Sensor A Circuit High - No sub type information	- Supercharger bypass valve actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Supercharger bypass valve actuator failure	- Refer to the electrical circuit diagrams and check supercharger bypass valve actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new supercharger bypass valve actuator as required
P25A2-00	Brake System Control Module Requested MIL Illumination - No sub type information	NOTE: **Monitor description.** Anti-lock brake system control module is indicating a fault to the engine control module that could have emission failure conditions therefore requests the MIL - Anti-lock brake system failure	- Check anti-lock brake system control module for additional DTCs and refer to relevant DTC index. Rectify these first - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		- High speed CAN bus failure - Fuse failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - Anti-lock brake system control module power circuit failure - Anti-lock brake system control module ground circuit failure	
P2601-00	Coolant Pump "A" Control Circuit performance / Stuck Off - No sub type information	- Connector is disconnected, connector pin is backed out, connector pin corrosion - Air charge coolant pump relay circuit short circuit to ground, short circuit to power, open circuit, high resistance - Air charge coolant pump relay failure	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check air charge coolant pump relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest - Check and install new air charge coolant pump relay as required
P2610-00	ECM/PCM Engine Off Timer Performance - No sub type information	NOTE: **Monitor description.** Engine control module monitors the global time and checks if it plausible - CAN network failure - Invalid CAN data received from central junction box	- Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required - Refer to the electrical circuit diagrams and check the power and ground connections to the central junction box - Check central junction box for related DTCs and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2610-84	ECM/PCM Engine Off Timer Performance - Signal below allowable range	NOTE: **Monitor description.** The engine	- Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first

		control module will set a DTC if the engine off time calculated from the global time in the instrument cluster/central junction box is less than 1hr, when the shutdown temperature was fully warm, and the coolant 1 temperature at start is significantly cooled down to ambient temp ▪ Vehicle battery has been isolated and the global time does not increment. Warm coolant has been replaced with cold coolant ▪ Rapid engine temperature cooling with rapid ambient temperature rise	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity for coolant temperature sensor and ambient temperature sensor ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2610-87	ECM/PCM Engine Off Timer Performance - Missing message	NOTE: **Monitor description.** Engine control module monitors the global time on CAN. If global time is not received then this DTC is set ▪ CAN network failure ▪ Central junction box not transmitting CAN data	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the central junction box ▪ Check central junction box for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2626-00	O2 Sensor Positive Current Trim Circuit/Open Bank 1 Sensor	▪ Heated oxygen sensor circuit, open circuit	▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for open circuit

	1 - No sub type information		
P2626-13	O2 Sensor Positive Current Trim Circuit / Open Bank 1 Sensor 1 - Circuit open	▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new heated oxygen sensor as required
P2629-00	O2 Sensor Positive Current Trim Circuit / Open Bank 2 Sensor 1 - No sub type information	▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ Check and install new heated oxygen sensor as required
P2635-7B	Fuel Pump A Low Flow / Performance - Low fluid level	▪ Fuel level too low ▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank ▪ Fuel pump driver module failure	▪ Check sufficient fuel is available ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed ▪ Check and install a new fuel pump driver module as required
P2635-92	Fuel Pump A Low Flow / Performance - Performance or incorrect operation	▪ The powertrain control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Fuel level too low	▪ Check sufficient fuel is available ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank ▪ Fuel pump driver module failure	▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed ▪ Check and install a new fuel pump driver module as required
P2635-97	Fuel Pump A Low Flow / Performance - Component or system operation obstructed or blocked	▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction ▪ Fuel level too low ▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank ▪ Fuel pump driver module failure	▪ Check sufficient fuel is available ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed ▪ Check and install a new fuel pump driver module as required
U0001-87	High Speed CAN Communication Bus - Missing message	▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ The powertrain control module has not received the expected CAN signal from the front audio control panel, button module within the specified time interval ▪ CAN harness link between powertrain control module and front audio control panel, button module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment, check front audio control panel, button module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check front audio control panel, button module power and ground circuits for open circuit. Check CAN harness between powertrain control module and front audio control panel, button module, repair as necessary

U0001-88	High Speed CAN Communication Bus - Bus off	- The engine control module has determined failures where a data bus is not available - High speed CAN bus failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - The engine control module has not received the expected CAN signal from the terrain response switchpack within the specified time interval - CAN harness link between engine control module and terrain response switchpack network malfunction	- Vehicle Conditions to enable DTC Logging strategy - Ignition on and engine running - Prioritised Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment carry out network integrity test - Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Using the Jaguar Land Rover approved diagnostic equipment, check terrain response switchpack for DTCs and refer to the relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check terrain response switchpack power and ground circuits for open circuit. Check CAN harness between engine control module and terrain response switchpack, repair as necessary
U0002-00	High Speed CAN Communication Bus Performance - No sub type information	**NOTE:** Engine control module will use default CAN signal values - High speed CAN short circuit to ground - High speed CAN short circuit to power	- Refer to the electrical circuit diagrams and check high speed CAN for short circuit to ground - Refer to the electrical circuit diagrams and check high speed CAN for short circuit to power
U0010-88	Medium Speed CAN Communication Bus - Bus off	- The engine control module has determined failures where a data bus is not available - Medium speed CAN bus failure - Medium speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit	- Using the Jaguar Land Rover approved diagnostic equipment carry out network integrity test - Refer to the electrical circuit diagrams and check medium speed CAN network for short circuit to ground, short circuit to power, open circuit

U0080-88	Vehicle Communication Bus F - Bus off	▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U0101-00	Lost Communication with TCM - No sub type information	▪ The engine control module has not received the expected CAN signal from the transmission control module within the specified time interval ▪ CAN harness link between engine control module and transmission control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check transmission control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transmission control module power and ground circuits for open circuit. Check CAN harness between engine control module and transmission control module, repair as necessary
U0102-00	Lost Communication with Transfer Case Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the transfer case control module within the specified time interval ▪ CAN harness link between engine control module and transfer case control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check transfer case control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transfer case control module power and ground circuits for open circuit. Check CAN harness between engine control module and transfer case control module, repair as necessary
U0103-00	Lost Communication With Gear Shift Control Module A - No sub type information	▪ The engine control module has not received the expected CAN signal from the transmission control switch within the specified time interval ▪ CAN harness link between engine control module and transmission control switch network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check transmission control switch for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check transmission control switch power and ground circuits for open circuit. Check CAN harness between engine control module and transmission control switch, repair as necessary
U0104-00	Lost Communication With Cruise	▪ The engine control module has not received the expected CAN signal from the speed	▪ Using the Jaguar Land Rover approved diagnostic equipment, check speed control module for DTCs and refer to the relevant DTC index

	Control Module - No sub type information	control module within the specified time interval ▪ CAN harness link between engine control module and speed control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check speed control module power and ground circuits for open circuit. Check CAN harness between engine control module and speed control module, repair as necessary
U0121-00	Lost Communication With Anti-Lock Brake System (ABS) Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti-lock brake system control module, repair as necessary
U0126-00	Lost Communication With Steering Angle Sensor Module - No sub type information	▪ Steering angle sensor module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Steering angle sensor system fault	▪ Refer to the electrical circuit diagrams and check the steering angle sensor module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the steering angle sensor module for related DTCs and refer to the relevant DTC index
U0126-88	Lost Communication With Steering Angle Sensor Module - Bus off	▪ The engine control module has not received the expected CAN signal from the steering angle sensor control module within the specified time interval	▪ Using the Jaguar Land Rover approved diagnostic equipment, check steering angle sensor control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check steering angle sensor control module

		▪ CAN harness link between engine control module and steering angle sensor control module network malfunction	power and ground circuits for open circuit. Check CAN harness between engine control module and steering angle sensor control module, repair as necessary
U0128-00	Lost Communication With Park Brake Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the parking brake control module within the specified time interval ▪ CAN harness link between engine control module and parking brake control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check parking brake control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check parking brake control module power and ground circuits for open circuit. Check CAN harness between engine control module and parking brake control module, repair as necessary
U012A-00	Lost Communication with Chassis Control Module "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the integrated suspension control module within the specified time interval ▪ CAN harness link between engine control module and integrated suspension control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check integrated suspension control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and integrated suspension control module, repair as necessary
U012B-00	Lost Communication with Chassis Control Module "B" - No sub type information	▪ The engine control module has not received the expected CAN signal from the integrated suspension control module within the specified time interval ▪ CAN harness link between engine control module and integrated suspension control module network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check integrated suspension control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and integrated suspension control module, repair as necessary
U0138-	Lost		

00	Communication with All Terrain Control Module - No sub type information	▪ Terrain response switchpack power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Terrain response system fault	▪ Refer to the electrical circuit diagrams and check the terrain response switchpack power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the terrain response switchpack for related DTCs and refer to the relevant DTC index
U0138-88	Lost Communication with All Terrain Control Module - Bus off	▪ The engine control module has not received the expected CAN signal from the terrain response switchpack within the specified time interval ▪ CAN harness link between engine control module and terrain response switchpack network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check terrain response switchpack for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check terrain response switchpack power and ground circuits for open circuit. Check CAN harness between engine control module and terrain response switchpack, repair as necessary
U0140-00	Lost Communication With Body Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the central junction box within the specified time interval ▪ CAN harness link between engine control module and central junction box network malfunction	▪ Using the Jaguar Land Rover approved diagnostic equipment, check central junction box for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check central junction box power and ground circuits for open circuit. Check CAN harness between engine control module and central junction box, repair as necessary
U0146-00	Lost Communication With Gateway "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the gateway control module within the specified time interval	▪ Using the Jaguar Land Rover approved diagnostic equipment, check gateway control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test

		■ CAN harness link between engine control module and gateway control module network malfunction	■ Refer to the electrical circuit diagrams and check gateway control module power and ground circuits for open circuit ■ Check CAN harness between engine control module and gateway control module, repair as necessary
U0151-00	Lost Communication With Restraints Control Module - No sub type information	■ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ■ CAN harness link between engine control module and restraints control module network malfunction	■ Using the Jaguar Land Rover approved diagnostic equipment, check restraints control module for DTCs and refer to the relevant DTC index ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check restraints control module power and ground circuits for open circuit. Check CAN harness between engine control module and restraints control module, repair as necessary
U0151-08	Lost Communication With Restraints Control Module - Bus Signal / Message Failures	■ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ■ CAN harness link between engine control module and restraints control module network malfunction	■ If this DTC is logged with U0151-00 & B10A2-32, check for fuse failure, restraints control module power and ground circuits for open circuit. Refer to the electrical circuit diagrams and check CAN harness ■ If this DTC is logged with B10A2-32, or on its own refer to the electrical circuit diagrams and check restraints control module crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the Jaguar Land Rover approved diagnostic equipment, check restraints control module for DTCs and refer to the relevant DTC index ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check restraints control module power and ground circuits for open circuit. Check CAN harness between engine control module and restraints control module, repair as necessary
U0155-00	Lost Communication With Instrument Panel Cluster (IPC) Control Module - No sub type information	■ The engine control module has not received the expected CAN signal from the instrument cluster within the specified time interval ■ CAN harness link between engine control module and	■ Using the Jaguar Land Rover approved diagnostic equipment, check instrument cluster for DTCs and refer to the relevant DTC index ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check instrument cluster power and ground circuits for open circuit. Check CAN harness between engine control module

		instrument cluster network malfunction	and instrument cluster, repair as necessary
U0164-00	Lost Communication With HVAC Control Module - No sub type information	▪ Automatic temperature control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Climate control system fault	▪ Refer to the electrical circuit diagrams and check the automatic temperature control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the automatic temperature control module for related DTCs and refer to the relevant DTC index
U0164-88	Lost Communication With HVAC Control Module - Bus off	▪ CAN harness link between engine control module and automatic temperature control module network malfunction ▪ The engine control module has not received the expected CAN signal from the automatic temperature control module within the specified time interval	▪ Using the Jaguar Land Rover approved diagnostic equipment, check automatic temperature control module for DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check automatic temperature control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and automatic temperature control module, repair as necessary
U0167-00	Lost Communication With Vehicle Immobilizer Control Module - No sub type information	▪ Power is lost from the engine control module or the central junction box during the immobilizer learn routine	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the engine control module and central junction box ▪ Using the Jaguar Land Rover approved diagnostic equipment, carry out the immobilisation procedure ▪ Check for CAN network interference /engine control module related errors
U025D-88	Lost Communication With Front Controls	▪ The engine control module has not received the expected CAN signal	▪ Using the Jaguar Land Rover approved diagnostic equipment, check rear integrated control panel for DTCs and refer to the relevant DTC index

	Interface Module "B" - Bus off	from the rear integrated control panel within the specified time interval ■ CAN harness link between engine control module and rear integrated control panel network malfunction	■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check rear integrated control panel power and ground circuits for open circuit. Check CAN harness between engine control module and rear integrated control panel, repair as necessary
U0284-07	Lost Communication with Active Grille Air Shutter Module A - Mechanical failures	■ Active grille air shutter mechanical failure ■ CAN harness link between powertrain control module and active grille air shutter module network malfunction ■ The powertrain control module has not received the expected CAN signal from the active grille air shutter module within the specified time interval	■ Check active grille air shutter for mechanical failure ■ Using the Jaguar Land Rover approved diagnostic equipment check active grille air shutter module for related DTCs and refer to the relevant DTC index ■ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter module ■ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ■ Check the CAN circuit between powertrain control module and active grille air shutter module. Repair as necessary
U0300-00	Internal Control Module Software Incompatibility - No sub type information	■ The engine control module has not received the expected master configuration data transmitted from the vehicle ■ Engine control module hardware part incorrect for application ■ Engine control module software part incorrect for application	■ Refer to the electrical circuit diagrams and check CAN network circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Check engine control module hardware part is correct for application ■ Check engine control module software part is correct for application
U0402-00	Invalid Data Received from TCM - No sub type information	■ Engine control module relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Stop/ start failure indicated to the engine control	■ Refer to the electrical circuit diagrams and check engine control module relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Check transmission control module for related DTCs and refer to relevant DTC index ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical

		module by the transmission control module	circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-02	Invalid Data Received from TCM - General signal failure	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal ▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on and engine running ▪ Prioritised Checks to Perform ▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-29	Invalid Data Received from TCM - Signal invalid	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal ▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-41	Invalid Data Received from TCM - General checksum failure	NOTE: **Monitor description.** Monitors transmission control module CAN data for erroneous alive counter, checksum, complement information ▪ CAN network failure	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required

		Implausible CAN data received from transmission control module	
U0402-62	Invalid Data Received from TCM - Signal compare failure	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal - CAN network failure - Implausible CAN data received from transmission control module	- Check transmission control module for related DTCs and refer to relevant DTC index - Check transmission control switch for related DTCs and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-67	Invalid Data Received from TCM - Signal incorrect after event	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal - CAN network failure - Implausible CAN data received from transmission control module	- Check transmission control module for related DTCs and refer to relevant DTC index - Check transmission control switch for related DTCs and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-81	Invalid Data Received from TCM - Invalid serial data received	NOTE: **Monitor description.** Transmission is indicating a fault to the engine management system that could have emission impact and requests the MIL	- Vehicle Conditions to enable DTC Logging strategy - Ignition on engine running - Prioritised Checks to Perform - Check transmission control module for related DTCs and refer to relevant DTC index - Check transmission control switch for related DTCs and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required

		▪ CAN network failure ▪ Implausible CAN data received from transmission control module	
U0402-82	Invalid Data Received from TCM - Alive / sequence counter incorrect / not updated	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal ▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0402-83	Invalid Data Received from TCM - Value of signal protection calculation incorrect	NOTE: **Monitor description.** Integrity check of transmission control module CAN signal ▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on engine running ▪ Prioritised Checks to Perform ▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required
U0404-00	Invalid Data Received from Gear Shift Control Module A - No sub type information	▪ CAN network failure ▪ Invalid CAN data received from transmission control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground

			connections to the transmission control switch ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0404-41	Invalid Data Received from Gear Shift Control Module A - General checksum failure	▪ CAN network failure ▪ Invalid CAN data received from transmission control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control switch ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0404-82	Invalid Data Received from Gear Shift Control Module A - Alive / sequence counter incorrect / not updated	▪ CAN network failure ▪ Invalid CAN data received from transmission control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control switch ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0404-83	Invalid Data Received from Gear Shift Control Module A - Value of signal protection calculation incorrect	▪ CAN network failure ▪ Invalid CAN data received from transmission control switch	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first

			- Check transmission control switch for related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control switch - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0405-68	Invalid Data Received From Cruise Control Module - Event information	- Speed control system failure - Speed control buttons jammed /contaminated /damaged - Clock spring failure	- Check speed control module for additional DTCs and refer to relevant DTC index - Check speed control buttons and clock spring are not jammed/contaminated /damaged - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0405-82	Invalid Data Received From Cruise Control Module - Alive / sequence counter incorrect / not updated	- Speed control system failure - Speed control buttons jammed /contaminated /damaged - Clock spring failure	- Check speed control module for additional DTCs and refer to relevant DTC index - Check speed control buttons and clock spring are not jammed/contaminated /damaged - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0405-84	Invalid Data Received From Cruise Control Module - Signal below allowable range	- Speed control system failure - Speed control buttons jammed /contaminated /damaged - Clock spring failure	- Check speed control module for additional DTCs and refer to relevant DTC index - Check speed control buttons and clock spring are not jammed/contaminated /damaged - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

			▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0405-86	Invalid Data Received From Cruise Control Module - Signal invalid	▪ Speed control system failure ▪ Speed control buttons jammed /contaminated /damaged ▪ Clock spring failure	▪ Check speed control module for additional DTCs and refer to relevant DTC index ▪ Check speed control buttons and clock spring are not jammed/contaminated /damaged ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0415-02	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - General signal failure	NOTE: **Monitor description.** The anti-lock brake system control module CAN messages are being updated on the CAN bus. If the data is not updated and correctly formatted periodically a DTC is set. The monitor is operational when the ignition is ON ▪ Other anti-lock brake system control module DTCs are set ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit	▪ Check anti-lock brake system control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit

		to ground, short circuit to power, open circuit - Anti-lock brake system control module power circuit failure - Anti-lock brake system control module ground circuit failure	
U0415-29	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Signal invalid	**NOTE:** **Monitor description.** The anti-lock brake system control module CAN messages are being updated on the CAN bus. If the data is not updated and correctly formatted periodically a DTC is set. The monitor is operational when the ignition is ON - High speed CAN bus failure - Fuse failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - Anti-lock brake system control module power circuit failure - Anti-lock brake system control module ground circuit failure	- Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0415-41	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module -	**NOTE:** **Monitor description.** The anti-lock	- Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test

	General checksum failure	brake system control module CAN messages are being updated on the CAN bus. If the data is not updated and correctly formatted periodically a DTC is set. The monitor is operational when the ignition is ON - High speed CAN bus failure - Fuse failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - Anti-lock brake system control module power circuit failure - Anti-lock brake system control module ground circuit failure	- Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0415-61	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Signal calculation failure	- All surface progress control system fault - Anti-lock brake system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check the terrain response switchpack for related DTCs and refer to the relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0415-62	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Signal compare failure	NOTE: **Monitor description.** Monitors the compliment of the anti-lock brake system control module CAN signal engine running request	- Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity

		■ Stop/ start system failure ■ Harness failure - Wiring integrity anti-lock brake system control module	
U0415-82	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Alive / sequence counter incorrect / not updated	NOTE: **Monitor description.** The anti-lock brake system control module CAN messages are being updated on the CAN bus. If the data is not updated and correctly formatted periodically a DTC is set. The monitor is operational when the ignition is ON ■ High speed CAN bus failure ■ Fuse failure ■ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ■ Anti-lock brake system control module power circuit failure ■ Anti-lock brake system control module ground circuit failure	■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ■ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ■ Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0415-83	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Value of signal protection calculation incorrect	NOTE: **Monitor description.** The anti-lock brake system control module CAN messages are being updated on the	■ Vehicle Conditions to enable DTC Logging strategy ■ Ignition on engine running ■ Prioritised Checks to Perform ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test

		CAN bus. If the data is not updated and correctly formatted periodically a DTC is set. The monitor is operational when the ignition is ON - High speed CAN bus failure - Fuse failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - Anti-lock brake system control module power circuit failure - Anti-lock brake system control module ground circuit failure	- Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0416-46	Invalid Data Received From Vehicle Dynamics Control Module - Calibration / parameter memory failure	- CAN network failure - Invalid CAN data received from anti-lock brake system control module	- Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required - Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first - Check anti-lock brake system control module for related DTCs and refer to relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the ‘Diagnosis Menu’ tab and retest
U0416-68	Invalid Data Received From Vehicle Dynamics Control Module - Event information	- The engine control module has received the default brake pressure signal value over CAN from the anti-lock brake system control	- Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for DTCs and refer to the relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment carry out network integrity test. Refer to the electrical circuit diagrams and check anti-lock

		module for a specified time interval ▪ Anti-lock brake system failure ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti-lock brake system control module, repair as necessary
U0417-41	Invalid Data Received From Park Brake Control Module - General checksum failure	▪ Other electric park brake control module DTCs are set ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Electric park brake control module power circuit failure ▪ Electric park brake control module ground circuit failure	▪ Check electric park brake control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Refer to the electrical circuit diagrams and check electric park brake control module power and ground circuits for open circuit
U0417-82	Invalid Data Received From Park Brake Control Module - Alive /sequence counter incorrect / not updated	▪ Other electric park brake control module DTCs are set ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Electric park brake control module power circuit failure ▪ Electric park brake control module ground circuit failure	▪ Check electric park brake control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Refer to the electrical circuit diagrams and check electric park brake control module power and ground circuits for open circuit
U0426-00	Invalid Data Received From Vehicle Immobilizer	▪ Electric steering column lock has received an invalid identity response ▪ Module substituted	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required. Using the Jaguar Land Rover approved

	Control Module - No sub type information		diagnostic equipment check and up-date the car configuration file as required ▪ Ensure all modules installed in the vehicle which store vehicle identity are valid for this vehicle and are not substitutes from a donor vehicle
U042B-29	Invalid Data Received from Chassis Control Module, A - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Fuse failure ▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Integrated suspension control module failure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Ignition on ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check integrated suspension control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U042B-68	Invalid Data Received from Chassis Control Module, A - Event information	▪ Fuse failure ▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Deployable spoiler failure	**NOTE:** Operational requirements needed to allow the monitor to be fully tested. Ignition on ▪ Check deployable spoiler for correct operation ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check integrated suspension control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for

			short circuit to ground, short circuit to power, open circuit - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U042B-82	Invalid Data Received from Chassis Control Module, A - Alive/sequence counter incorrect/not updated	- Fuse failure - High speed CAN bus failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit - Integrated suspension control module failure	NOTE: Operational requirements needed to allow the monitor to be fully tested. Ignition on - Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit - Check integrated suspension control module for additional DTCs and refer to relevant DTC index. Rectify these first - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test - Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0447-00	Invalid Data Received From Gateway "A" - No sub type information	NOTE: **Monitor description.** Engine control module has been informed of a failure within the dual battery system - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity dual battery system	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Harness failure - Wiring integrity gateway module	
U0452-00	Invalid Data Received From Restraints Control Module - No sub type information	NOTE: **Monitor description.** Engine control module has been informed of a failure within the seat belt sensor ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity seat belt sensor ▪ Harness failure - Wiring integrity gateway module	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U0592-00	Invalid Data Received From Gear Shift Control Module B - No sub type information	▪ CAN network failure ▪ Invalid CAN data received from transmission control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Check transmission control switch for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control switch ▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U1A14-00	CAN Initialisation Failure - No sub type information	▪ Harness fault - CAN circuit ▪ Engine control module failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Ignition on engine running ▪ Prioritised Checks to Perform

			▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new engine control module as required
U2012-00	Car Configuration Parameter(s) - No sub type information	NOTE: **Monitor description.** Car configuration parameter received from central junction box is outside the designated limits allowed ▪ Car configuration signal not received ▪ Car configuration file incorrect ▪ Harness fault - CAN circuit ▪ Central junction box not transmitting some or all of the car configuration CAN data	▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required ▪ Check the central junction box for related DTCs and refer to the relevant DTC index ▪ Clear the DTC and re-test
U2012-02	Car Configuration Parameter(s) - General signal failure	▪ The final drive ratio is output as part of the vehicle/car configuration. If the value received by the powertrain control module does not match the internally used value this DTC will be set ▪ Car configuration signal not received ▪ Car configuration file incorrect ▪ Harness fault - CAN circuit	▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the high speed CAN control modules with the latest level software ▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required ▪ Check the body control module for related DTCs and refer to the relevant DTC index ▪ Clear the DTC and re-test

U2108-00	Adaptive Cruise Control - No sub type information	- Adaptive speed control system failure - Error indicating adaptive speed control failure flag set	- Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-24	Adaptive Cruise Control - Signal stuck high	- The engine control module measures a signal that remains high when transitions are expected - Adaptive speed control system failure - Adaptive speed control follow speed error	- Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-64	Adaptive Cruise Control - Signal plausibility failure	- Adaptive speed control system failure - Adaptive speed control follow speed range error	- Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-68	Adaptive Cruise Control - Event information	- Adaptive speed control system failure - Error indicating adaptive speed control follow speed check when stationary	- Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
U2108-86	Adaptive Cruise Control - Signal invalid	- Adaptive speed control system failure - Error when invalid adaptive speed control resume requests are present	- Check adaptive speed control module for DTCs and refer to the relevant DTC index. Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required